



Skolkovo Institute of Science and Technology

MASTER'S THESIS

**Data-Driven Indoor Occupancy Monitoring Using CO2 Levels:
Enhancing Energy Efficiency and Comfort**

Master's Educational Program: Center for Energy Science and Technology

Student: _____

Hamza Said

Center for Energy Science and Technology

Research Advisor: _____

Elena Gryazina

Assistant Professor

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МАГИСТЕРСКАЯ ДИССЕРТАЦИЯ

Мониторинг заполненности помещений на основе данных об уровне CO₂: Повышение энергоэффективности и комфорта

Магистерская образовательная программа: Центр Энергетических Наук и
Технологий

Студент: _____

Хамза саид

Центр Энергетических Наук и Технологий

Научный руководитель: _____

Елена Грязина

Старший Преподаватель

Москва 2025

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Data-Driven Indoor Occupancy Monitoring Using CO₂ Levels: Enhancing Energy Efficiency and Comfort

Hamza Said

Skolkovo Institute of Science and Technology

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Abstract

This thesis presents a data-driven approach to indoor occupancy detection based on carbon dioxide (CO₂) concentration measurements. CO₂ sensors were deployed in residential and academic environments, including dormitories, classrooms, and meeting rooms, to collect continuous environmental data. The primary objective was to investigate the relationship between CO₂ levels and human presence, and to develop machine learning models capable of accurately predicting occupancy.

Multiple models were explored, ranging from classical algorithms (Random Forest, k-NN, XGBoost) to deep learning architectures, with a focus on Long Short-Term Memory (LSTM) networks. The results demonstrate that incorporating temporal features from CO₂, temperature, and humidity data significantly enhances the accuracy of occupancy classification. Notably, the LSTM model achieved a 97% accuracy and F1-score in multi-class classification without requiring data augmentation or resampling techniques.

This research provides a robust, non-intrusive solution for real-time occupancy monitoring and paves the way for intelligent HVAC control. By leveraging CO₂ dynamics as a proxy for human presence, the proposed method contributes to optimizing indoor air quality and energy efficiency in modern building systems.

Research Advisor:

Name: Elena Gryazina

Degree: PhD

Title: Assistant Professor

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1. Introduction

1.1. Energy Consumption in Buildings

Over the past decade, global primary energy consumption has shown a steady upward trend, with the building and construction sector emerging as a dominant contributor to both national and international energy demand. Buildings, encompassing residential, commercial, and industrial spaces, are a cornerstone of modern infrastructure but also a significant driver of energy use. In developed nations, the building sector alone accounts for an estimated 20--40% of primary energy consumption [1], underscoring its substantial role in shaping energy policies and sustainability goals.

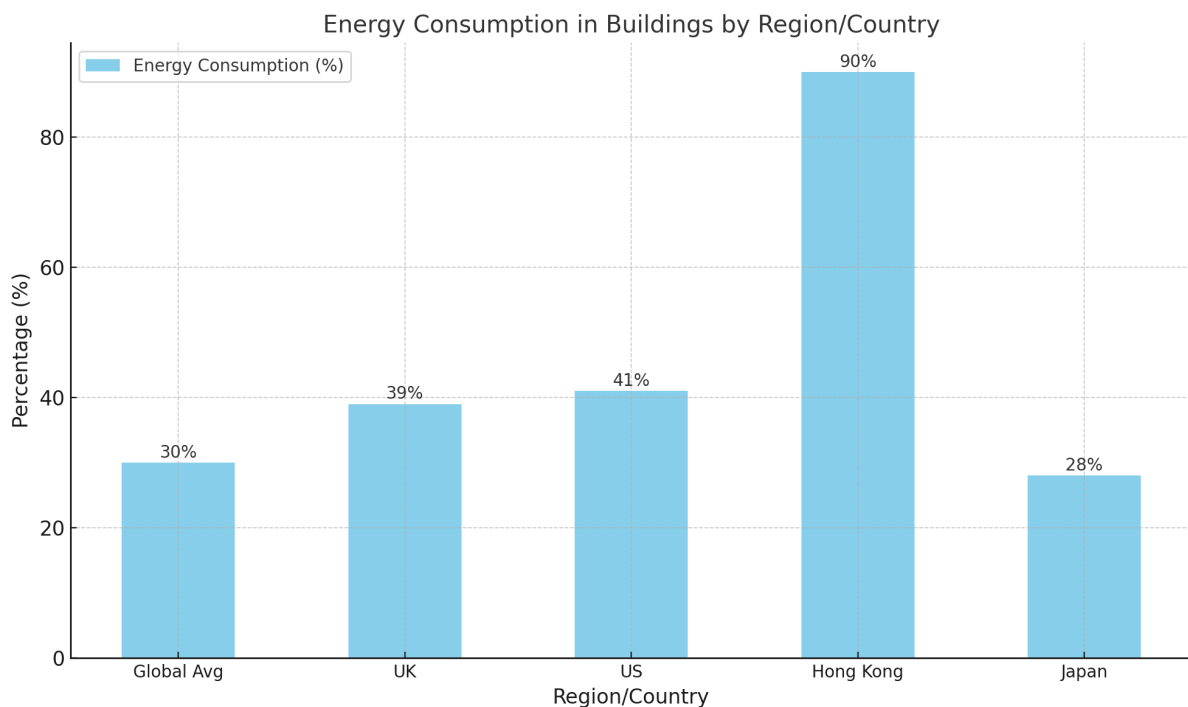


Рис. 1: Energy consumption in buildings by region/country. The figure highlights the percentage of total energy consumption attributed to buildings in different regions, with Hong Kong leading at 90%, followed by the United States and the United Kingdom.

For example, in the United Kingdom, energy consumption attributable to buildings surpasses that of many other European nations by approximately 39%, exceeding the continental average of 37% [1]. This trend illustrates the energy-intensive nature of building operations, which include heating, cooling, lighting, and powering appliances. In the United States, the residential and commercial building sectors collectively consume about 41% of the nation's total energy. A striking 74% of this consumption is specifically allocated to electricity, highlighting the heavy reliance of modern buildings on electrical systems to maintain comfort and functionality [2, 3].

Similarly, in Hong Kong, the building sector's energy footprint is even more pronounced. Approximately 90% of the region's total energy supply is consumed by buildings, contributing to 61% of total greenhouse gas emissions [2, 3]. This dual impact of high energy consumption

and significant emissions positions Hong Kong's building sector as a critical area for energy efficiency interventions and climate change mitigation strategies.

In Japan, the building sector constitutes 28% of the country's total energy consumption, with residential and commercial buildings accounting for 14.8% and 13.2%, respectively. This significant share reflects the growing demand for energy-intensive technologies and the increasing urbanization of the country. According to the International Energy Agency (IEA), the construction industry in Japan emerged as the largest consumer of energy in 1999 [3, 4]. This highlights not only the scale of energy use within the sector but also its critical importance in shaping Japan's energy policy and sustainability framework.

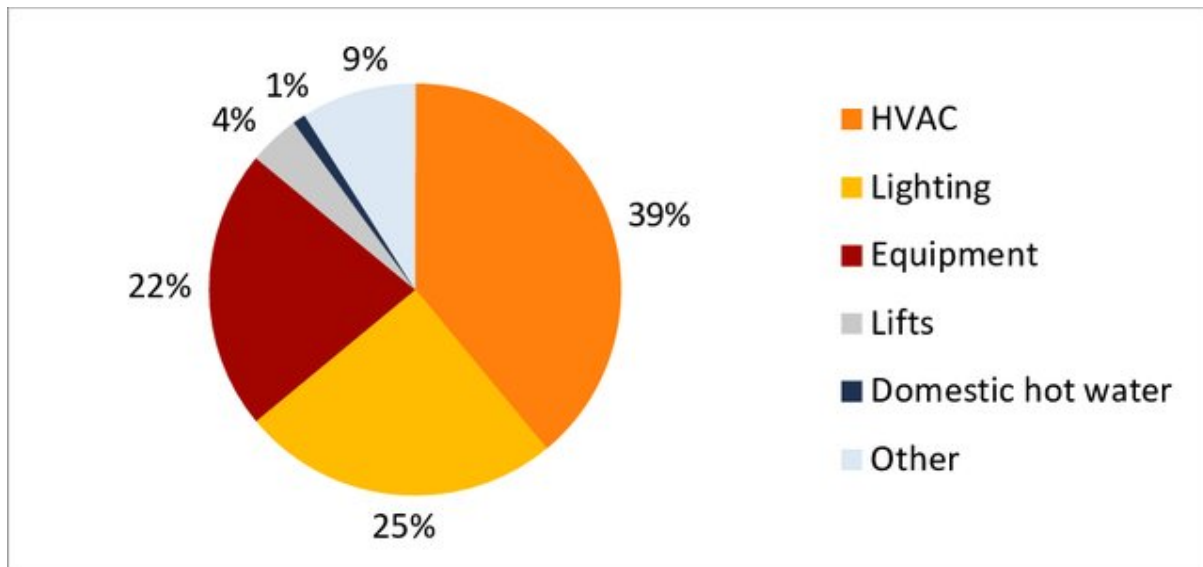


Рис. 2: Typical energy consumption breakdown within an office building. HVAC systems dominate the energy profile, followed by lighting and equipment, highlighting key areas for energy efficiency improvements [5].

The pie chart in Figure 2 illustrates the typical energy consumption breakdown within an office building, highlighting the significant contribution of various systems to total energy use. HVAC systems dominate the energy consumption profile, accounting for 39%, reflecting their critical role in maintaining thermal comfort and air quality. Lighting follows at 25%, underscoring the impact of illumination needs on energy usage, while equipment accounts for 22%, representing the energy demands of electronic devices and operational tools. Other contributors include lifts (4%), domestic hot water (9%), and miscellaneous systems (1%). These figures emphasize the importance of targeting HVAC systems and lighting for energy efficiency improvements, as they collectively account for a majority of the energy consumed in office buildings. This breakdown provides valuable insights into energy management priorities and supports efforts to optimize building operations for sustainability and cost savings [5].

These statistics collectively underline the pivotal role of buildings in global energy consumption. With urbanization on the rise and technological advancements increasing energy demand, the building sector is set to remain a major player in energy-related challenges. Addressing energy efficiency in buildings is therefore imperative to mitigate environmental impacts, reduce operational costs, and achieve broader sustainability goals.

1.2. Importance of Indoor Occupancy and Behavior

The International Energy Agency–Energy in Buildings and Communities (IEA-EBC), Annex 53, identifies six key factors that significantly impact the overall performance of buildings. These factors include the building envelope, climatic conditions, energy and service systems, interior design specifications, building operation and maintenance practices, and occupant behavior [6]. Current research on building energy savings predominantly focuses on operational energy, emphasizing aspects such as energy and service systems, as well as building operation and maintenance [7, 8].

In recent years, however, there has been a growing recognition of occupant behavior (OB) as a cost-effective approach to reducing energy consumption in buildings. OB refers to the actions taken by building occupants that influence energy use, including controlling lighting, HVAC (heating, ventilation, and air conditioning) systems, windows, blinds, and other energy-related building features [9]. Despite its importance, accurately predicting occupant behavior remains a significant challenge due to the diverse and individualistic nature of human actions, which inherently vary among different occupants [10].

1.3. Building Performance Simulation and Occupant Behavior

The building performance simulation (BPS) platform is an effective and cost-efficient option for analyzing and improving building design and energy management, while precise input of occupant behavior is fundamentally important [11, 12]. Additionally, BPS can be used during the operational stage to assess real building performance and diagnose building systems that may not be functioning correctly. If refurbishment or remodeling is required during the maintenance stage, BPS can identify the most energy-efficient retrofit model [13]. In other words, BPS assists with fault detection in HVAC operation and forecasts potential energy savings associated with system-level modifications, building redesign, or retrofits to enhance the building's energy performance [10, 14].

However, a significant limitation of existing BPS tools under realistic conditions is the inaccurate or misleading input parameters related to occupant behavior and building operation [14]. Another constraint is the inability of BPS to simulate realistic building performance [15]. Achieving accurate and reliable outcomes requires simulating buildings under realistic conditions, including occupant behavior (OB) [14, 16]. In general, OB is primarily represented by the settings for indoor temperature, scheduling of equipment, lighting, and HVAC systems [17, 18]. These parameters are highly variable and inherently unpredictable for individual occupants or groups of occupants [19, 20]. They also have a significant impact on real energy consumption and overall building energy performance [7, 21].

Nowadays, BPS incorporating occupant behavior has the ability to predict near-real energy consumption [22]. Improving the understanding of occupant behavior is therefore essential for assessing its influence on the overall performance of the building [23]. To understand the energy–behavior interaction, recent studies in residential and commercial buildings have shown that occupant behavior within indoor environments has a dual impact on both total building energy consumption and occupants' comfort with the quality of the indoor environment [24]. Given this energy–comfort connection, researchers can no longer afford to treat occupant-related factors as fixed boundary conditions. Default assumptions about occupant behavior in energy modeling are increasingly being replaced by dynamic modeling approaches. Consequently, recent simulation studies have focused on various methods for simulating occupant behavior,

with several approaches being applied to mimic occupants' conduct in building energy research [25].

1.4. Relevance of CO₂ Standards and Air Quality

Indoor CO₂ concentrations serve as a critical marker for air quality, influencing occupant comfort, health, and productivity. Adhering to established CO₂ standards ensures that indoor spaces remain conducive to both human well-being and optimal building performance. Figure 3 illustrates these thresholds based on international, regional, and national standards.

1.4.1. International Guidelines

Low CO₂ Levels (350–800 ppm): These levels are equivalent to outdoor air and are generally considered safe. They align with recommendations by the American Society of Heating, Refrigerating, and Air-Conditioning Engineers (ASHRAE) [26] and the Building Hygiene Control Law in Japan.

Moderate CO₂ Levels (800–1000 ppm): This range signifies acceptable healthy levels for normal indoor air quality. Prolonged exposure beyond this threshold may result in mild discomfort and reduced efficiency.

Elevated CO₂ Levels (1000–2500 ppm): Concentrations within this range can cause drowsiness, discomfort, and noticeable odors. Ventilation is essential to maintain air quality and ensure occupant comfort.

High CO₂ Levels (2500–5000 ppm): Sustained exposure to these levels poses significant health risks, including respiratory problems and cognitive impairment. Emergency interventions and improved ventilation systems are critical at this stage.

1.4.2. Russian Standards for CO₂ Concentration

The Russian guidelines for public and residential indoor air quality provide the following thresholds:

- **Up to 1000 ppm:** Optimal levels that ensure occupant comfort and acceptable air quality.
- **1000–1400 ppm:** Allowable for short durations; ventilation adjustments are advised to avoid prolonged exposure.
- **Above 1400 ppm:** These levels necessitate immediate ventilation improvements to restore air quality and mitigate health risks.

These standards underscore the importance of effective ventilation systems in maintaining acceptable air quality, as outlined in Russian building codes [27].

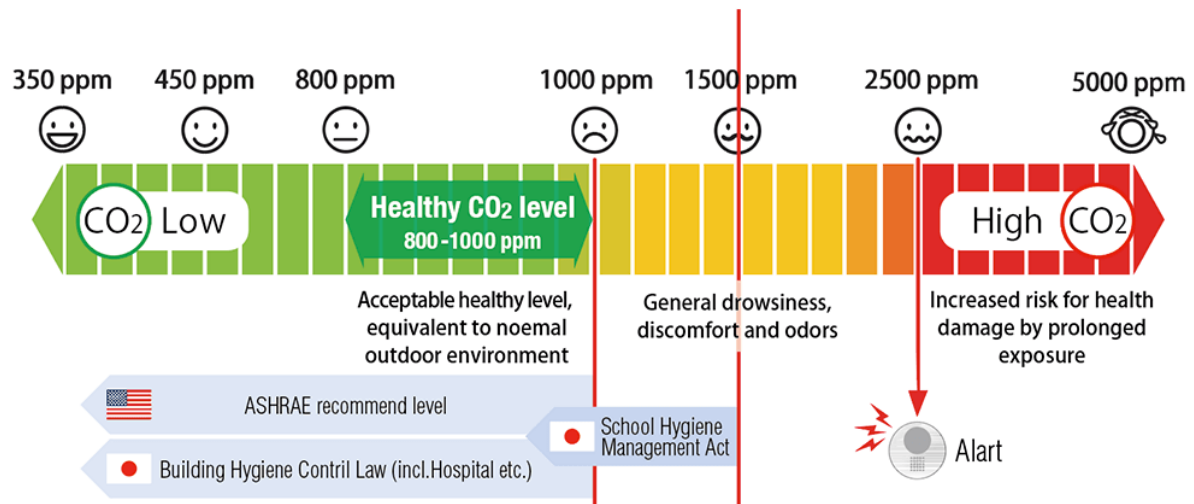


Рис. 3: CO₂ concentration levels and their effects on health and comfort. The diagram[28] highlights thresholds for low, moderate, and high CO₂ levels, referencing ASHRAE, Japan, and Russian standards.

1.5. Importance of Indoor Comfort and Occupant Well-Being

In addition to energy efficiency, indoor comfort and health play a pivotal role in building performance. The concept of indoor comfort encompasses thermal, acoustic, visual, and air quality conditions that collectively influence the well-being and productivity of occupants. Poorly maintained indoor environments can lead to discomfort, reduced cognitive performance, and even long-term health issues. Factors such as inadequate ventilation, extreme temperatures, and poor air quality have been linked to conditions like respiratory problems and fatigue, underscoring the importance of maintaining optimal indoor conditions[29].

1.6. Problem Statement

Residential buildings represent a substantial portion of global energy consumption, with heating, ventilation, and air conditioning (HVAC) systems being major contributors. These systems often operate on fixed schedules or simplistic rule-based triggers that do not account for real-time human presence. This disconnect can lead to energy inefficiency and poor indoor air quality (IAQ), especially when ventilation is insufficient during high occupancy periods.

Indoor carbon dioxide (CO₂) levels, primarily driven by human respiration, offer a reliable proxy for estimating occupancy. However, accurately inferring the number of occupants from CO₂ and other environmental signals remains a challenging task—particularly in residential settings, where behavior is highly variable and sensor deployment is limited.

To illustrate this challenge, Figure 4 shows CO₂ concentration levels recorded at 5-minute intervals in the author’s dormitory room during the month of January. The red dashed line marks the ASHRAE-recommended comfort threshold of 1000 ppm. As shown, CO₂ levels frequently exceed this limit, indicating prolonged periods of poor ventilation and highlighting the need for more responsive, occupancy-informed building systems.

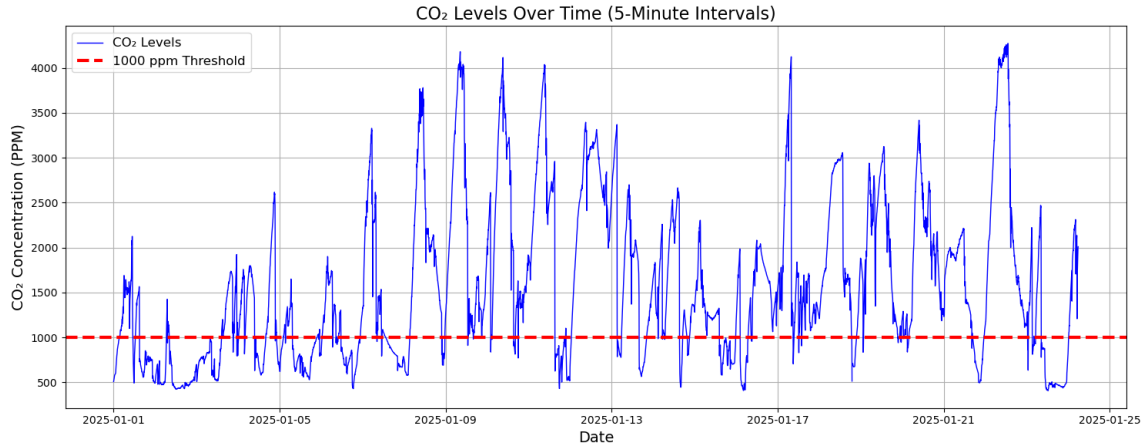


Рис. 4: CO₂ concentration over time during January in the author's dormitory room, recorded at 5-minute intervals. The red dashed line indicates the ASHRAE-recommended comfort threshold (1000 ppm). Frequent and prolonged exceedances emphasize the potential of CO₂-based occupancy estimation in improving future ventilation strategies.

This study addresses the problem of occupancy estimation by developing a data-driven model that predicts the number of people in a room based on CO₂, temperature, and humidity sensor data. Accurate and timely occupancy estimation is a foundational step toward enabling intelligent HVAC control strategies that can reduce energy use while maintaining thermal comfort and air quality in residential environments.

1.7. Objectives

The primary objective of this thesis is to develop a comprehensive and reliable method for detecting indoor occupancy, which can be utilized to optimize energy efficiency and maintain high indoor air quality (IAQ) standards in buildings. Given the increasing energy demands of the building sector and the significant role of occupant behavior in energy dynamics, this research aims to bridge existing gaps in occupancy detection technologies and their application in building management systems. The detailed objectives are as follows:

1. To conduct a thorough analysis of energy consumption trends in the building sector, emphasizing the influence of occupancy patterns on energy use and the resulting implications for indoor air quality and thermal comfort. This objective will highlight the critical need for accurate and real-time occupancy data in improving building performance.
2. To review and critically evaluate existing methods for indoor occupancy detection, such as those relying on CO₂ sensors, motion detectors, and other environmental monitoring tools. The limitations of these methods in terms of scalability, precision, and integration with modern building systems will be systematically identified.
3. To design a novel method for real-time occupancy detection, leveraging sensor data, such as CO₂ concentrations, temperature, humidity, and other environmental parameters. This method will aim to enhance the accuracy of detecting occupant numbers while addressing challenges related to dynamic and unpredictable human behavior.
4. To develop a framework for integrating the proposed occupancy detection method into HVAC (heating, ventilation, and air conditioning) systems. This framework will focus

on enabling dynamic control of ventilation and thermal conditions based on real-time occupancy data, thereby reducing energy consumption and improving IAQ.

5. To validate the proposed occupancy detection method through experimental studies conducted in real-world settings, such as dormitories, classrooms, and other shared spaces. Data collected from these environments will be used to assess the method's reliability and effectiveness in diverse scenarios.
6. To explore the broader implications of the proposed method for sustainable building management practices. This includes evaluating the scalability of the solution for integration into smart building systems and its potential contribution to energy policies aimed at reducing greenhouse gas emissions.
7. To provide actionable recommendations for future research and practical implementation of occupancy detection technologies, with a focus on enhancing their applicability in both existing and newly constructed buildings.

2. Introduction to CO_2 Dynamics

2.0.1. Mass Balance Principle

Understanding the behavior of indoor CO_2 concentration requires analyzing the balance between sources, sinks, and the overall dynamics of air exchange within a defined space. Indoor CO_2 levels are influenced by external factors, such as ventilation and ambient air quality, as well as internal factors, such as the number of occupants and their activity levels [30, 31]. A mathematical representation of these dynamics provides insights into transient and steady-state CO_2 levels, enabling non-intrusive estimation of human occupancy and optimization of ventilation systems for energy efficiency and comfort [32, 33].

The mass balance principle applies to the conservation of mass within a controlled system. In the context of indoor environments, it represents the dynamic behavior of CO_2 concentration in a closed or semi-closed space. This principle can be expressed as the change in mass being equal to the input rate minus the output rate, plus the generation rate. For indoor CO_2 dynamics:

- The input rate refers to fresh outdoor air entering the room.
- The output rate accounts for exhausted air removing CO_2 .
- The generation rate corresponds to the CO_2 produced by occupants.

The rate of change of indoor CO_2 concentration over time can be described mathematically. Let $C_{CO_2}(t)$ represent the indoor CO_2 concentration at time t , with an initial concentration of $C_{CO_2,0}$. The ambient outdoor CO_2 concentration is C_{ext} , while r denotes the per capita CO_2 generation rate, n is the number of occupants, $\dot{m}$ is the air exchange rate, and V is the room's volume. The governing equation is:

$$\frac{dC_{CO_2}}{dt} = \frac{\dot{m}}{V}(C_{ext} - C_{CO_2}) + \frac{r \cdot n}{V}. \quad (1)$$

This equation captures how ventilation and human activity jointly influence CO_2 levels, providing a foundation for further analysis and modeling.

Step-by-Step Solution:

1. Rearrange terms to simplify the equation:

$$\frac{dC_{\text{CO}_2}}{dt} + \frac{\dot{m}}{V}C_{\text{CO}_2} = \frac{\dot{m}}{V}C_{\text{ext}} + \frac{r \cdot n}{V}. \quad (2)$$

2. Recognize it as a first-order linear differential equation. Solve using integrating factors:

$$C_{\text{CO}_2}(t) = C_{\text{CO}_2,0}e^{-\frac{\dot{m}}{V}t} + \left(C_{\text{ext}} + \frac{r \cdot n}{\dot{m}}\right) \left(1 - e^{-\frac{\dot{m}}{V}t}\right). \quad (3)$$

3. Literature Review

3.1. Overview of Occupancy Estimation Methods

Occupancy estimation is a critical component of energy-efficient building management and indoor environmental control. Accurate knowledge of occupant presence and count allows heating, ventilation, and air conditioning (HVAC) systems to adapt dynamically, reducing energy waste while maintaining comfort and air quality. Over the past two decades, a range of methods has been developed to estimate occupancy using either predefined schedules, sensor-based measurements, or intelligent data analysis. These approaches can be grouped into four major categories:

- **Simulation-Based Methods:** These methods rely on predefined occupant schedules or agent-based models implemented within Building Performance Simulation (BPS) tools. While useful for scenario analysis and design-phase planning, their accuracy is limited during operation due to static assumptions and lack of real-time feedback.
- **Physics-Based Models:** These models use first-principles equations, such as mass-balance formulations of CO₂ concentration, to infer occupancy levels from known parameters like ventilation rate and room volume. They offer strong interpretability and low data requirements but are sensitive to environmental variability and require precise inputs.
- **Physics-Informed Statistical Models:** These are hybrid models that embed physical knowledge (e.g., CO₂ dynamics) into statistical inference frameworks like Hidden Markov Models (HMMs) or auto-regressive processes. Notably, the Switching Auto-Regressive Hidden Markov Model (Switching AR-HMM) introduces time-dependent probabilistic reasoning while retaining physical interpretability, enabling robust estimation even without labeled data.
- **Data-Driven Approaches:** These approaches leverage supervised machine learning or deep learning models trained on historical sensor data (e.g., CO₂, temperature, humidity, Wi-Fi usage) to learn patterns associated with occupancy. While often achieving high accuracy, they require significant labeled data and may lack transparency.

Each category offers distinct advantages and limitations depending on deployment context, building type, data availability, and required granularity of occupancy information. In the following subsections, we review representative works in each category, highlighting their methodologies, performance, and suitability for real-time occupancy detection in smart building systems.

3.2. Simulation-Based Approaches

Simulation-based methods estimate occupancy using predefined schedules or synthetic behavioral profiles embedded within building performance simulation (BPS) tools. These approaches are commonly applied in the design and planning phases of buildings to forecast energy demand, ventilation needs, and indoor comfort levels under assumed occupancy patterns [34, 35].

Typical simulation platforms such as EnergyPlus, TRNSYS, and IDA-ICE [36, 37] integrate standard occupancy schedules derived from building codes, surveys, or regulatory guidelines. Agent-based models [38] extend this by simulating individual or group behavior based on probabilistic movement and activity patterns.

Although valuable for design-phase evaluation, these models suffer from key limitations when applied to real-time control:

- **Static Assumptions:** Fixed occupancy schedules fail to capture short-term variability in occupant behavior [34].
- **No Sensor Feedback:** These models do not adapt to real-time environmental data or unexpected usage patterns [35].
- **Lack of Personalization:** Generalized behavior models often neglect spatial, cultural, or temporal heterogeneity in occupant routines.

Despite these challenges, simulation-based methods remain essential for benchmarking, sensitivity analysis, and early-phase HVAC system sizing. However, for real-time adaptive control, more responsive sensor-based or hybrid approaches are typically required.

3.3. Physics-Based Approaches

Physics-based approaches to occupancy estimation rely on first-principles modeling of physical processes that are affected by human presence—most notably, the buildup of carbon dioxide (CO₂) due to human respiration. These models typically employ mass-balance equations to relate the concentration of CO₂ in a space to parameters such as occupancy count, ventilation rate, room volume, and ambient CO₂ levels [39, 40].

The core of this method is the dynamic CO₂ mass-balance equation, usually expressed as:

$$\frac{dC(t)}{dt} = \frac{G(t)}{V} - \frac{Q}{V}(C(t) - C_{out})$$

where $C(t)$ is the indoor CO₂ concentration at time t , $G(t)$ is the CO₂ generation rate (proportional to the number of occupants), V is the room volume, Q is the ventilation rate, and C_{out} is the outdoor CO₂ level.

By solving or inverting this equation, one can estimate $G(t)$, and hence occupancy, provided the other parameters are known or estimated. These models are attractive due to their transparency and reliance on interpretable parameters. They are especially useful in naturally ventilated buildings, where changes in CO₂ are more directly influenced by occupancy events.

However, physics-based models face several practical challenges:

- **Parameter Sensitivity:** Accurate occupancy estimation depends on knowing or estimating ventilation rates, which may vary with time, window state, or HVAC operations [41].
- **Assumption of Well-Mixed Air:** Most models assume perfect mixing of indoor air, which is rarely true in real-world scenarios, especially in larger or partitioned rooms [40].

- **Offline Use and Calibration Needs:** These models often require offline calibration or manual tuning to reflect the actual building environment [42].

Despite these limitations, physics-based approaches serve as a fundamental layer for more advanced hybrid and learning-based models. Their deterministic nature makes them highly interpretable and suitable for model validation or digital twin implementations, particularly in early-stage design or scenarios where labeled data is scarce.

3.4. Physics-Informed Statistical Models

To address the limitations of purely physics-based or data-driven methods, researchers have explored hybrid approaches that incorporate domain knowledge into probabilistic frameworks. These **physics-informed statistical models** integrate first-principles understanding—such as the CO₂ mass-balance equation—into inference mechanisms like Hidden Markov Models (HMMs) or state-space models, enabling interpretable yet flexible estimation of occupancy patterns.

A notable example in this category is the **Switching Autoregressive Hidden Markov Model (Switching AR-HMM)**, proposed by Esmayeeli-Sikaroudi et al. [43]. This approach models indoor CO₂ concentration dynamics using a system of autoregressive equations governed by hidden states, each corresponding to a specific occupancy regime (e.g., 0, 1, or 2+ occupants). The transitions between these hidden states are governed by a Markov process, while the CO₂ evolution within each state follows a physically grounded mass-balance formulation. Crucially, the method is trained in an *unsupervised* fashion—requiring only raw CO₂ time series data and no labeled occupancy information.

Key Advantages.

- **Physics-Driven Interpretability:** The use of mass-balance equations ensures that the inferred occupancy regimes have physical meaning, which improves transparency and diagnostic potential.
- **No Labeled Data Needed:** The model can infer occupancy from unlabeled CO₂ data, making it scalable to real-world deployments where manual labeling is impractical.
- **Noise Robustness:** Through the use of temporal filtering and autoregressive dynamics, the model smooths noisy CO₂ patterns and handles transient events more effectively than static classifiers.
- **Ventilation Rate Learning:** The method simultaneously learns both the occupancy regime transitions and the associated ventilation parameters, improving adaptability to building-specific conditions.

Limitations. Despite these strengths, the Switching AR-HMM and related models have several constraints:

- **Manual Regime Configuration:** The number of hidden states must be defined a priori, which can introduce bias and may require empirical tuning across different environments.
- **Assumes Well-Mixed Air:** The approach assumes single-zone, well-mixed air conditions, which may not hold in naturally ventilated or partitioned spaces with airflow stratification.

- **CO₂ Dominance Required:** The model assumes that CO₂ dynamics are primarily driven by occupancy. If ventilation noise or window operation dominates the signal, estimation accuracy can degrade.
- **Not Real-Time Ready:** The method is designed for batch training on historical data. Adapting it for real-time inference or feedback control remains an open challenge.

Physics-informed statistical models offer a compelling middle ground between mechanistic interpretability and data-driven flexibility. They are especially suitable for settings where labeled data is unavailable but CO₂ dynamics are strong indicators of human presence. However, their practical adoption requires careful calibration and may be constrained in multi-zone or dynamic ventilation environments.

3.5. CO₂-Based Data-Driven Approaches

Data-driven methods for occupancy estimation have rapidly evolved due to the increasing availability of low-cost environmental sensors and the rise of powerful machine learning (ML) techniques. Among these, the use of CO₂ concentration as a proxy signal has become especially prominent, owing to its strong correlation with human metabolic activity in enclosed indoor spaces. When people exhale, they release CO₂ into the air; hence, fluctuations in indoor CO₂ levels can serve as a reliable indicator of changes in occupancy. However, translating raw CO₂ readings into accurate occupancy predictions presents a number of challenges related to sensor noise, ventilation variability, room geometry, and temporal lags in gas mixing.

Overview of Methods and Inputs. Researchers have adopted a variety of supervised and unsupervised learning techniques to model the relationship between CO₂ concentration (and its derivatives) and occupancy. Table 1 provides a non-exhaustive summary of prior work, listing the input parameters used, models applied, and performance metrics. Most studies rely on a core set of features: raw CO₂ levels, rate of change (first derivative), and moving averages over 5, 10, or 15-minute windows to smooth short-term fluctuations. Additional environmental variables such as temperature, humidity, air pressure, and light intensity are often included to improve robustness and reduce false positives.

Traditional Machine Learning Models. Early efforts focused on shallow models like k-Nearest Neighbors (kNN), Linear Discriminant Analysis (LDA), Decision Trees, and Support Vector Machines (SVM) [44, 45, 46]. These models were computationally efficient and interpretable, making them attractive for real-time applications. However, they often struggled with temporal dependencies inherent in CO₂ dynamics, as well as nonlinear interactions between variables. For instance, Candanedo et al. [45] compared LDA, CART, Random Forest (RF), and Gradient Boosted Machines (GBM) using CO₂, humidity, light, and temperature data. While performance was generally satisfactory in single-zone test cases, models did not generalize well to different buildings without retraining.

Advanced Tree-Based and Ensemble Methods. To improve prediction accuracy and model resilience, more recent studies have used ensemble methods like Random Forests, Gradient Boosted Trees (GB), and AdaBoost [47, 48]. These models aggregate multiple weak learners to capture complex interactions and reduce overfitting. Brennan et al. [47], for example, trained RF and GB models on Raspberry Pi-collected environmental data, showing improved performance

in classifying low, medium, and high occupancy states. These methods also allowed for feature importance analysis, revealing that CO₂ variation and temperature were consistently strong predictors.

Neural Networks and Deep Learning. As datasets grew in size and diversity, deep learning models began to show superior performance in handling complex, nonlinear mappings between sensor inputs and occupancy targets. Artificial Neural Networks (ANNs), Multi-Layer Perceptrons (MLPs), and Extreme Learning Machines (ELMs) were among the first to be adopted [49, 50]. More recently, Recurrent Neural Networks (RNNs) such as Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), and their bidirectional variants have been employed to capture time-dependent patterns in CO₂ evolution [51]. LSTMs, in particular, are capable of modeling long-range dependencies between past CO₂ dynamics and current occupancy state, which static models cannot do.

For example, Tekler et al. [51] developed a benchmark framework comparing traditional models (SVM, RF, kNN) with deep recurrent models (LSTM, Bi-LSTM, GRU, Bi-GRU). Using a rich dataset comprising CO₂, energy usage, temperature, and Wi-Fi probe requests across multiple zones, they found that deep learning models outperformed classical ML across all metrics, particularly in multi-class occupancy classification tasks. However, they also noted the computational cost and the risk of overfitting in small datasets.

Hybrid Data Sources and Multi-Modal Sensing. While CO₂ is often the core predictor, several studies have integrated complementary data streams to boost performance. Wang et al. [52, 53] and Hobson et al. [54] incorporated Wi-Fi probe data, passive infrared (PIR) sensor readings, sound levels, and plug-load energy usage alongside CO₂ to create hybrid input vectors. These multimodal setups significantly improved accuracy, especially in transient conditions like short visits or low-occupancy edge cases. However, such setups come with increased cost, complexity, and potential privacy concerns, especially with camera or network-level data.

Feature Engineering and Preprocessing. Preprocessing is crucial to extracting useful signals from raw sensor data. Many studies reported the use of:

- **Temporal Smoothing:** Moving averages or exponential smoothing to remove noise and align sensor lag with actual occupancy changes.
- **Normalization:** Z-score or min-max normalization to account for inter-room or inter-building variability.
- **CO₂ Derivatives:** First-order or second-order changes in CO₂ to capture entering/leaving patterns.
- **Windowed Aggregation:** Statistical descriptors (mean, std, max, skewness) computed over sliding windows (5–30 min) for dynamic modeling.

Challenges and Limitations. Despite promising results, CO₂-based data-driven approaches face several limitations:

- **Need for Labeled Data:** Most supervised models require high-quality labeled occupancy data, which is labor-intensive to collect.

- **Generalization Issues:** Models trained in one building often fail to generalize across buildings with different ventilation setups or occupancy habits.
- **Sensor Drift and Calibration:** CO₂ sensors can exhibit long-term drift, requiring periodic calibration or signal correction.
- **Latency and Mixing Delays:** There is a time lag between changes in occupancy and measurable changes in CO₂, particularly in well-ventilated or large spaces.
- **Complex Interactions:** Ventilation systems, open windows, and outdoor CO₂ levels can introduce noise unrelated to occupancy, affecting prediction reliability.

Summary. CO₂-based data-driven models provide a scalable, non-intrusive means of estimating occupancy, particularly when privacy, cost, and deployment ease are key considerations. When combined with proper feature engineering and modern ML techniques like LSTM or Random Forests, these models can yield high prediction accuracy in various indoor settings. However, attention must be paid to data availability, generalizability, and sensor reliability for robust deployment in real-world HVAC control systems.

3.6. Occupancy Detection in Residential Buildings

While occupancy detection has been extensively studied in commercial and institutional environments such as offices, classrooms, and conference rooms, its application to residential buildings remains underexplored. The majority of published studies focus on relatively predictable occupancy patterns observed in workplaces, where well-defined operating hours and structured activities simplify modeling. However, residential environments present unique challenges due to their diverse occupant behaviors, irregular schedules, and more variable spatial and ventilation configurations.

Challenges Unique to Residential Settings. Unlike commercial buildings, residential apartments exhibit high temporal variability in occupancy, with patterns influenced by individual lifestyles, cultural habits, household size, and personal preferences. Daytime occupancy may vary significantly due to remote work, vacations, or stay-at-home members. Additionally, residential zones are typically smaller and less ventilated, which amplifies the effect of short-term CO₂ buildup and makes sensor placement critical. Moreover, residential units often have natural ventilation (e.g., openable windows), creating unpredictable CO₂ dissipation dynamics.

From a sensing perspective, deploying multiple sensors in homes is constrained by cost, aesthetics, and privacy concerns. Residents may resist intrusive sensors (e.g., cameras, PIR) or frequent battery replacements for IoT devices. These practical limitations make CO₂-based estimation attractive but also demand higher model robustness under noisy or sparse data.

Sparse Literature Coverage. Few studies have explicitly targeted residential occupancy detection. Tekler et al. [51] evaluated deep learning models for occupancy detection across multiple zones, including one residential-style setting. However, the primary focus remained on public buildings. Similarly, Hobson et al. [54] briefly discussed home occupancy but emphasized the need for more controlled environments. Zuraimi et al. [55] identified limitations in generalizing results from office-based CO₂ models to residential scenarios.

Candanedo et al. [45], while not focusing solely on homes, observed poor transferability of models trained in one environment to another. This observation supports the hypothesis

that residential settings require dedicated modeling pipelines and locally tuned algorithms. In a related vein, Wang et al. [52] used a laboratory resembling a residential room but emphasized the difficulty of collecting ground truth labels without violating privacy.

Need for Domain-Specific Models. The evidence above underscores the need to tailor occupancy estimation methods specifically for residential buildings. Factors that merit attention include:

- **Non-repetitive Routines:** Unlike office settings with structured patterns, residents follow varied daily routines influenced by work-from-home schedules, social habits, and lifestyle choices.
- **Multifunctional Spaces:** Rooms in residential units often serve multiple purposes (e.g., bedrooms doubling as study spaces), complicating zone-level detection.
- **Sensor Deployment Constraints:** Aesthetic and privacy limitations may limit the number and type of sensors that can be installed.
- **Limited Ground Truth Data:** Validating occupancy in homes is more challenging due to ethical and logistical constraints in using cameras or conducting surveys.

These factors challenge the direct application of models trained in institutional settings to residential ones. Moreover, they highlight the importance of adaptive, self-correcting algorithms that can infer local behavior patterns over time.

Таблица 1: Previous research on the CO₂ concentration based on the occupancy estimation.

Input Data	ML Model	Accuracy Factors	References
CO ₂ , temperature, humidity, pressure	ELM	Accuracy	[49]
CO ₂ , temperature, humidity, pressure	ELM, SVM, ANN, LDA, kNN, CART	Accuracy, NRMSE	[44]
CO ₂ , temperature, humidity	GB, kNN, LDA, RF	Accuracy, RMSE, NRMSE, CV	[47]
CO ₂	ELM	RMSE	[56]
CO ₂ , 15 min CO ₂ concentration mean, indoor-outdoor CO ₂ ratio	CART, HMM	RMSE, Accuracy	[57]
CO ₂ , CO ₂ variation, 5 min CO ₂ mean	SVM, ANN, PEM	RMSE, Fb	[55]
CO ₂	GCForest, SVM, CART, IHMM	EA, MAE, DA	[58]
CO ₂ , temperature, humidity, light	LDA, CART, RF, GBM	Accuracy	[45]
CO ₂ , temperature, light, PIR, sound	LDA, QDA, SVM, RF	Accuracy, F1 score	[46]
CO ₂ , temperature, humidity, pressure, TVOC, sound, window state	FNN	Accuracy	[50]
CO ₂ , light energy consumption, temperature, humidity	LDA, RF, CART	Accuracy	[59]
CO ₂ , temperature, humidity, Wi-Fi probe	kNN, SVM, ANN	MAE, RMSE, MAPE	[52]
CO ₂ , Wi-Fi probe, PIR, energy consumption	ANN	R ² , NRMSE	[54]
Environmental sensing, camera, Wi-Fi probe	ANN	MAE, F1 score	[53]
Energy consumption	RF, SVM, kNN, ANN, GB	Accuracy, Precision	[60]
CO ₂	GCForest, HMM	EA, DA	[61]
Indoor and outdoor environmental sensing, Wi-Fi probe, energy consumption	DNN, LSTM, Bi-LSTM, GRU, Bi-GRU	-	[51]
CO ₂ , temperature, humidity, dew point	SVM, AdB, RF, GB, LR, MLP	RMSE, MAE, MAPE, R ²	[48]
CO ₂ , temperature, humidity, air pressure	CDBLSTM	Accuracy, NRMSE	[62]

Masood et al. [49] applied the extreme learning machine (ELM) model to predict CO₂ concentration, indoor temperature, humidity, and absolute pressure. Similarly, Chen et al. [44] explored several machine learning models, including ELM, support vector machine (SVM), artificial neural network (ANN), linear discriminant analysis (LDA), k-nearest neighbors (KNN), and classification and regression tree (CART), using similar input variables. Both studies classified occupancy into three levels: low (0–5 occupants), medium (6–10 occupants), and high (11–15 occupants), emphasizing the significance of occupancy estimation for understanding indoor environmental conditions.

Brennan et al. [47] used gradient boosting (GB), KNN, LDA, and random forest (RF) models to estimate CO₂ concentration, temperature, and humidity. Their experiments utilized sensors

mounted on a Raspberry Pi board, and due to limited data, occupancy was categorized into ranges for comparative analysis. Similarly, Jiang et al. [56] tackled challenges like irregular airflow and transient CO₂ spikes caused by occupant activities by smoothing the data to enhance its reliability for occupancy estimation. Ryu et al. [57] extended this by using 15-minute averages of CO₂ concentration, its variations, and the indoor-to-outdoor CO₂ ratio with CART to provide more stable inputs for occupancy predictions despite dynamic airflows.

Zuraimi et al. [55] addressed rapid CO₂ fluctuations by averaging concentrations over five minutes, resulting in more consistent data. Zhou et al. [58] utilized GCForest, SVM, CART, and IHMM models to analyze CO₂ concentration data, reducing noise to improve prediction accuracy. They identified cases where elevated CO₂ levels were not directly correlated with occupancy due to noise or irregular variations, emphasizing the importance of refined input variables.

Candanedo et al. [45] compared the performance of LDA, CART, RF, and gradient boosted model (GBM) using combinations of CO₂, temperature, humidity, and illumination data. Singh et al. [46] incorporated additional variables, including passive infrared (PIR), noise sensors, and lighting energy consumption, alongside CO₂ and temperature, to enhance accuracy using LDA, QDA, SVM, and RF models. Adeogun et al. [50] employed the FNN model to predict CO₂ concentration, temperature, humidity, pressure, total VOCs, and window state. They validated their models across different rooms, highlighting challenges in generalization.

Elkhokhi et al. [59] utilized LDA, RF, and CART models to estimate occupancy by analyzing CO₂, lighting energy consumption, temperature, and humidity. While energy consumption was a valuable input variable, it required additional preparation, such as the installation of smart plugs, to be measured accurately. The study excluded energy consumption in rooms with mechanical ventilation but aimed to integrate it in future experiments.

Wang et al. [52] used CO₂, temperature, humidity, and Wi-Fi connection data for occupancy estimation in a laboratory setting, modeling the data at five-minute intervals during working hours. Hobson et al. [54] employed ANN to estimate occupancy by combining CO₂ concentration, Wi-Fi connections, PIR, and energy consumption data. They observed a significant improvement in accuracy when Wi-Fi connection data was included. Wang et al. [53] also integrated environmental and Wi-Fi data, although privacy concerns and localization accuracy presented challenges that required careful handling. Their findings underscored the potential of Wi-Fi data for occupancy estimation while identifying areas for improvement.

These studies collectively showcase diverse methods for estimating occupancy based on CO₂ concentration and other environmental variables. They underscore the importance of selecting appropriate input variables, addressing noise and irregularities in CO₂ data, and validating models across different environments to achieve reliable predictions.

Concluding Remarks. The reviewed literature highlights a growing interest in occupancy detection for smart building applications, with particular emphasis on CO₂-based estimation due to its non-intrusive and cost-effective nature. While various modeling techniques exist—ranging from physical equations to deep neural networks—their performance and applicability often hinge on deployment context, data availability, and environmental factors. However, residential settings remain underexplored, especially in terms of robust, generalizable models that can operate under noisy, unlabeled, or naturally ventilated conditions. This thesis aims to fill this gap by developing and evaluating advanced machine learning techniques, specifically tailored to occupancy detection in residential buildings.

4. Experimental Setup

4.1. Sensor Deployment and Locations

Four sensors were strategically deployed in both academic and residential settings to capture environmental data. Satellite images of the buildings where the sensors are installed provide context for the study.



Рис. 5: Satellite images of the locations where sensors were installed. (Left) Skoltech campus, including Professor Elena Gryazina's office and the meeting room of the Center for Energy Systems. (Right) Residential buildings in Moscow, including Apartment 1 (Palisad) and Apartment 2.

4.2. Academic Settings

- **Professor Elena Gryazina's Office:** Located in the Skoltech campus, this office represents a low-occupancy academic environment. The sensor is placed centrally within the office to minimize external airflow interference.
- **Meeting Room (Center for Energy Systems):** This shared space hosts periodic meetings, resulting in fluctuating occupancy. The sensor is installed to monitor CO₂ variations during these events.

4.3. Residential Settings

- **Apartment 1 (Palisad):** A residence with four occupants, chosen to analyze medium-occupancy scenarios in a typical residential environment.
- **Apartment 2:** A low-occupancy apartment housing two residents, representing a controlled scenario with minimal variability.

4.4. Sensor Specifications and Placement

The environmental sensors used in this study include the CM1106 for CO₂ concentration and the Si7020 for temperature and relative humidity. The CM1106 operates based on the non-dispersive infrared (NDIR) principle, which measures CO₂ by analyzing the absorption of infrared light at a specific wavelength. The Si7020 sensor employs capacitive sensing technology for humidity measurement and a bandgap temperature sensor for precise temperature readings. Additionally, these sensors provide data on particulate matter (dust particles), enabling a more comprehensive assessment of indoor air quality.

The sensors record real-time data at one-minute intervals, capturing CO₂ concentration, temperature, humidity, and dust particle levels. This high-frequency data acquisition ensures the detection of dynamic environmental changes, which is essential for correlating occupancy patterns with indoor air quality metrics.

The sensors were placed on the floor in their respective locations to maintain uniformity in data collection and minimize measurement bias caused by elevated placement. Two sensors were installed at Skoltech: one in Professor Elena Gryazina's office and another in the meeting room of the Center for Energy Systems. The remaining two sensors were deployed in residential apartments. One sensor was installed in an apartment housing four occupants (Palisad), while the other was placed in an apartment with two occupants.

To ensure consistent and accurate measurements, the sensors were calibrated regularly during the data collection period. Their strategic placement and advanced capabilities facilitated reliable monitoring of indoor environmental dynamics, which is critical for validating the mass balance equation and occupancy estimation models.



Рис. 6: The CM1106 and Si7020 sensors used in the study. These sensors provide minute-by-minute readings of CO₂, temperature, humidity, and dust particle concentrations.

4.5. Satellite Image Context

The satellite images in Figure 5 illustrate the buildings where the sensors were installed. These visuals help contextualize the experimental setup by highlighting the geographic and structural differences between the academic and residential environments.

4.6. Ground Truth and Environmental Control

To validate the collected data, manual occupant counts were conducted periodically. Additionally:

- Windows were kept closed throughout the data collection process to minimize interference from external air. However, doors were operated as usual, being opened and closed by occupants as they entered and exited the rooms. This approach ensured that the recorded environmental data accurately reflected indoor conditions influenced by typical occupant behavior.
- HVAC systems operated under standard conditions to reflect typical indoor environments.

5. Methodology

This section presents the step-by-step procedures used to process the collected sensor data and develop accurate occupancy estimation models based on CO₂ and environmental parameters. It includes preprocessing, theoretical validation through a mass balance framework, and the design of machine learning algorithms tailored to infer indoor occupancy levels.

5.1. Overview

The overarching goal of this research is to determine if CO₂ concentration, alongside temperature, humidity, and door status, can reliably predict indoor occupancy. Following the completion of sensor deployment and data collection (described in Section 4), this methodology focuses on processing the acquired data. The process involves:

1. Environmental signal analysis and feature engineering to interpret and structure raw sensor data for modeling
2. Validation of observed CO₂ dynamics using a mass balance equation
3. Development and training of machine learning models

These steps ensure the development of a robust, non-intrusive, and real-time occupancy detection system.

5.2. Environmental Signal Analysis and Feature Engineering

The raw sensor data collected across all test environments consisted of high-frequency time series of CO₂ concentration, temperature, relative humidity, and dust levels. These data streams were subject to several quality issues typical of real-world deployments, such as missing values, noise, and outliers. A structured preprocessing pipeline was implemented to ensure that the data fed into the machine learning models was both accurate and representative of true environmental conditions.

Outliers, defined as extreme values lying outside normal operating ranges, were detected and removed using Z-score thresholding. Specifically, measurements exceeding three standard deviations from a rolling mean were flagged as anomalous and excluded. To reduce high-frequency noise caused by transient air disturbances, a moving average filter was applied to CO₂, humidity, and temperature signals. Gaps caused by occasional network disruptions were addressed through linear interpolation, ensuring continuity of temporal structure. Finally, all numerical features were normalized using min-max scaling to bring them into a consistent value range and avoid bias during training.

Beyond basic cleaning, substantial effort was devoted to extracting meaningful features that capture the dynamics of occupant-driven environmental change. Time-windowed statistics, such as rolling means and standard deviations, were computed to encode short-term trends. Additionally, first-order derivatives of CO₂ and temperature were calculated to reflect the rate of change, as rising or falling trends often signal occupancy transitions. Categorical indicators such as hour-of-day and door open/close events were also included to provide contextual cues.

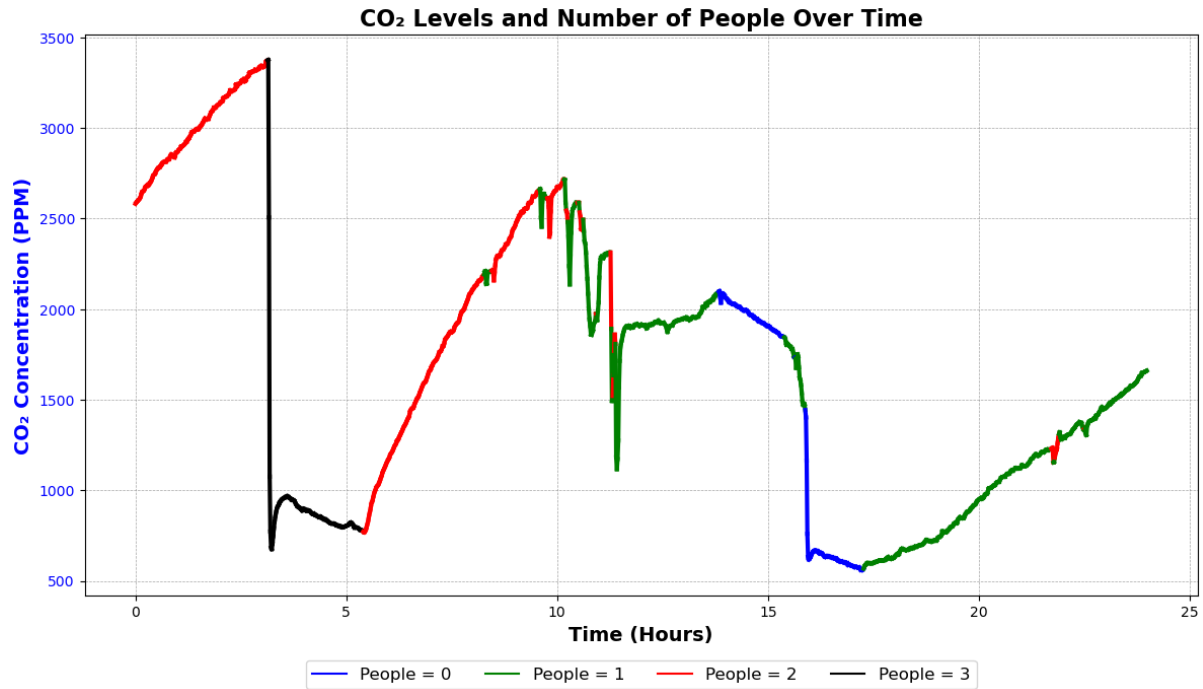


Рис. 7: CO₂ concentration (ppm) across a 24-hour period, segmented by occupant count. Blue represents unoccupied (0), green indicates 1 person, red shows 2 people, and black corresponds to 3 occupants..

Figure 7 presents the CO₂ concentration profile over a 24-hour window, with each segment color-coded by the number of people present. Several important behavioral patterns are observable in the signal. During occupied periods—especially when two or more people are present (red and black segments)—CO₂ tends to accumulate in a predictable, upward trajectory, reaching concentrations well above 3000 ppm.

Around the third hour, a sudden drop in CO₂ occurs despite continued occupancy (three people), suggesting a ventilation event such as a door or window opening. This illustrates how air exchange can override the expected accumulation trend even when the room remains occupied. Similar sharp declines occur multiple times between hours 9 and 11, where steep reductions in CO₂ levels interrupt ongoing occupancy (green and red segments). These discontinuities highlight how exogenous factors like door status or wind-driven ventilation affect the signal independent of human activity.

In contrast, the extended interval between hours 14 and 15.5 shows a gradual and smooth decay in CO₂, corresponding to an unoccupied room. This natural exponential-like decline matches the expected behavior of CO₂ dissipation in a closed space without mechanical ventilation. The distinction between abrupt and gradual declines is critical for feature engineering: sudden drops can mask occupancy if not accounted for, while slow decays are reliable indicators of room vacancy.

These observations collectively demonstrate that while CO₂ is a strong indicator of human presence, its dynamics are also influenced by room ventilation behavior. As such, the model must account for both rate-of-change and contextual events like door opening. To address this, features such as CO₂ first derivatives, ventilation-sensitive anomaly detection, and binary door state were incorporated in the learning pipeline. Additionally, smoothing and outlier detection became essential preprocessing steps to differentiate meaningful occupancy signals from short-term disturbances.

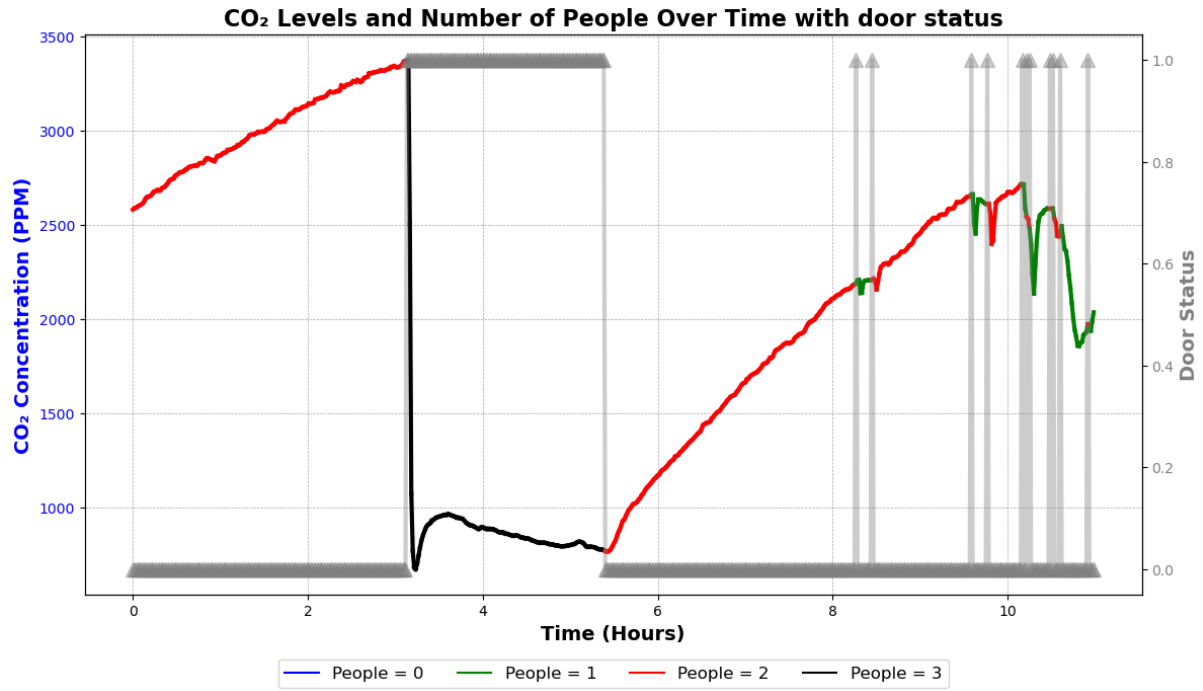


Рис. 8: CO₂ concentration over the first 10 hours, with vertical grey markers indicating door opening events. Blue, green, red, and black represent 0, 1, 2, and 3 occupants respectively.

Figure 8 provides a focused view of the first 10 hours of the day, where door status is overlaid on the CO₂ time series. Grey vertical arrows represent door opening events. This visualization allows us to isolate the effect of ventilation on CO₂ concentration during occupied periods. Several of the sharp dips in CO₂ levels — particularly around hours 3.2, 3.9, 6.0, and between 9 and 10 — align precisely with door opening events. This confirms that even brief ventilation can lead to rapid air exchange and a corresponding decline in CO₂, despite ongoing occupancy.

Importantly, the door status signal provides context for interpreting these disturbances. Without this additional variable, models could misinterpret abrupt CO₂ drops as a sign of room vacancy. Including door status as a binary or categorical feature helps disambiguate whether a CO₂ decline was driven by occupant departure or a ventilation event. This is particularly critical in time-series learning models, which are sensitive to transient anomalies in sequential sensor data.

The combination of these environmental and contextual cues reinforces the importance of multimodal sensing in non-intrusive occupancy detection. CO₂ alone may suggest presence or absence, but when paired with door activity, the signal becomes far more interpretable and robust. Consequently, door status was incorporated into the feature set not only for real-time modeling but also for validating and filtering derived CO₂ features such as slope, second derivative, and rolling window trends.

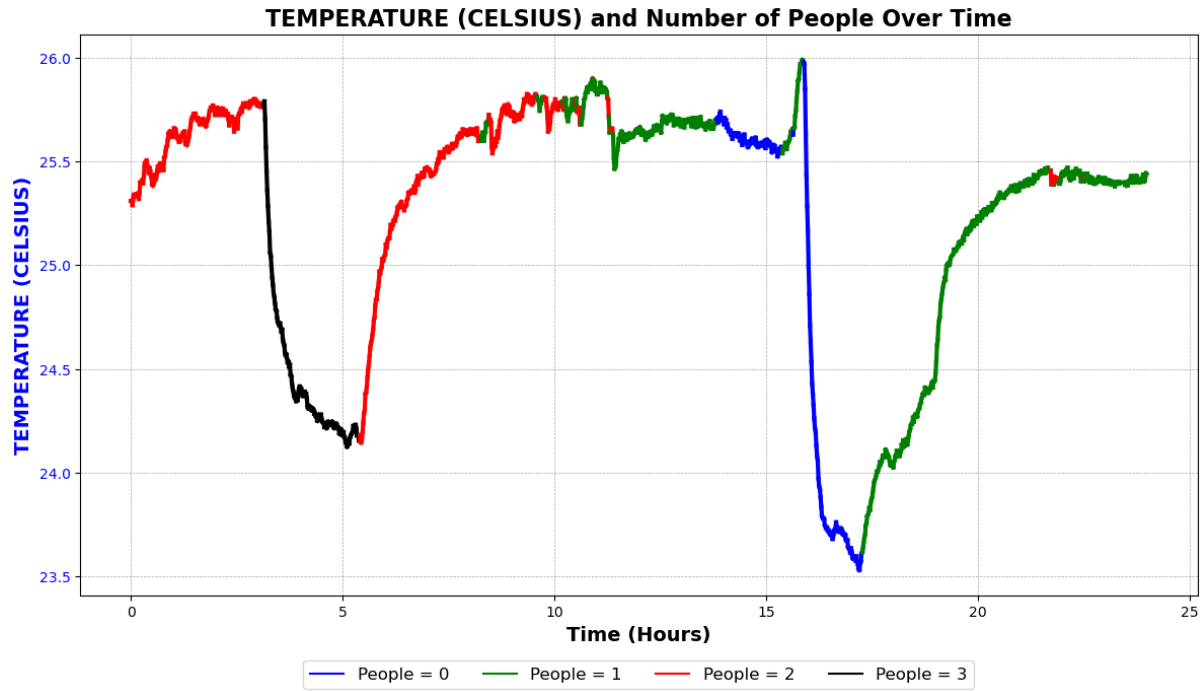


Рис. 9: Temperature (°C) across a 24-hour period, segmented by occupant count. Blue, green, red, and black represent 0, 1, 2, and 3 occupants respectively.

Figure 9 presents the 24-hour profile of indoor temperature, color-coded by occupancy levels. A strong positive correlation between occupant count and temperature is evident. During periods with two or more people, a consistent upward thermal trend is observed due to cumulative body heat and metabolic activity. The response is particularly pronounced during prolonged high-occupancy periods.

In addition to occupancy-driven changes, the temperature signal also exhibits sharp declines, some of which align with known door opening events or inferred ventilation moments. For example, sudden dips in temperature during otherwise continuous occupancy indicate external air intrusion or HVAC-triggered cooling. This behavior parallels the CO₂ signal, where ventilation can override accumulation patterns.

These observations reinforce that temperature is not only occupancy-sensitive, but also ventilation-reactive. As such, it was treated as both a primary occupancy feature and a complementary ventilation indicator. Derivative and rolling statistical features were engineered to capture both gradual heating patterns and abrupt drops, enabling the model to learn thermal shifts linked to human activity and environmental disturbances.

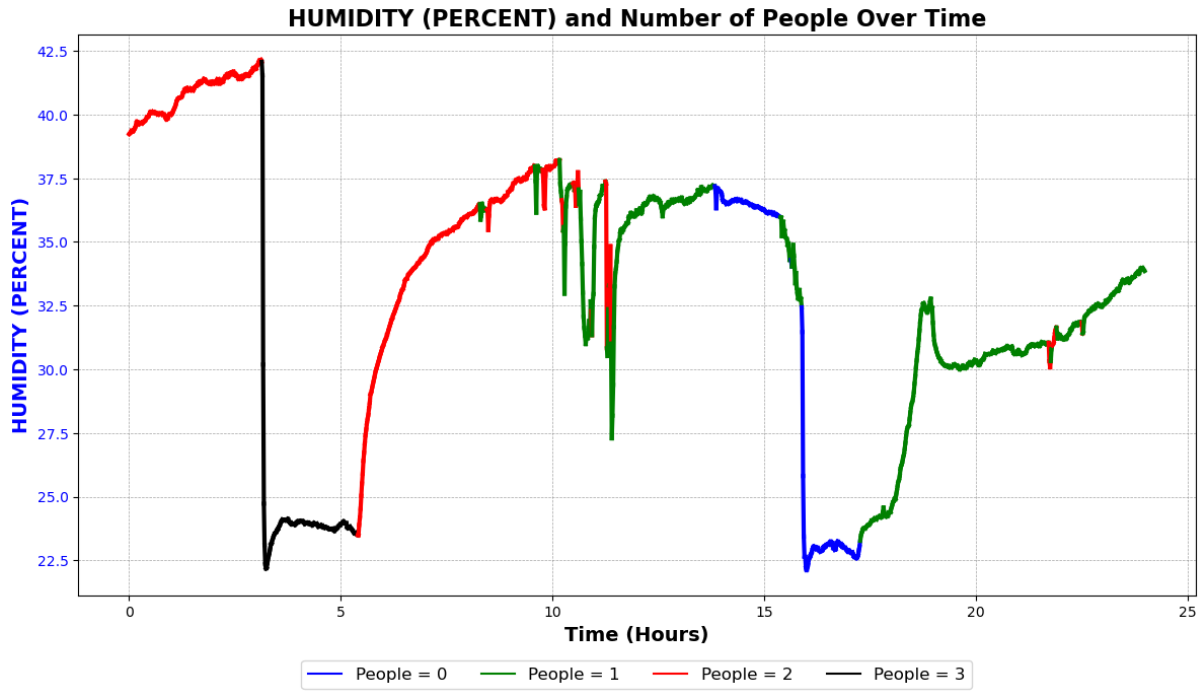


Рис. 10: Relative humidity (%) across a 24-hour period, segmented by occupant count. Higher humidity levels generally correlate with increased occupancy and show responsiveness to door-related ventilation events.

Figure 10 illustrates the variation in relative humidity throughout the same day, segmented by the number of occupants. As expected, humidity levels rise during periods of human presence due to respiratory moisture and indoor activity. This trend is especially apparent when occupancy increases from one to two or more individuals.

Beyond the direct correlation with occupancy, humidity also demonstrates sensitivity to door events. Several sharp declines in humidity levels occur during transitions that likely coincide with door openings or air exchange. These patterns closely resemble those observed in the CO₂ signal, where ventilation events cause temporary dips even during continued occupancy. For example, between hours 3 and 6, abrupt drops occur despite high occupancy levels, strongly indicating transient ventilation effects.

Although humidity tends to be more volatile and subject to short bursts of variability, its reactivity to both presence and ventilation makes it a valuable complementary feature. After applying noise smoothing and temporal aggregation, humidity-derived statistics were integrated into the model to help reinforce predictions where CO₂ or temperature alone might be ambiguous.

These combined signal patterns confirm that humidity, like CO₂ and temperature, encodes occupancy behavior while also responding to door status. Their joint modeling provides a more robust foundation for non-intrusive, sensor-driven inference.

In summary, each of the environmental signals—CO₂, temperature, and humidity—was not only examined for its direct correlation with occupancy but also assessed for responsiveness to ventilation-related events, particularly door openings. These dual behavioral signatures informed our feature extraction pipeline. Temporal patterns such as rising trends during occupancy, sudden dips during ventilation, and smooth decays during vacancy were translated into meaningful features. Specifically, we computed first and second-order derivatives, rolling window statistics (means and standard deviations), and binary markers such as door open/closed status. These features allowed the machine learning models to capture not just static environmental conditions,

but also their underlying dynamics—enabling reliable, real-time estimation of indoor occupancy under varying environmental conditions.

5.3. Theoretical Validation Using CO₂ Mass Balance Equation

To theoretically support the use of CO₂ as an indicator of occupancy, the mass balance model was used. This equation models the CO₂ concentration in a closed space as a function of external CO₂ levels, number of occupants, and ventilation rate. The equation is given by:

$$C_{CO_2}(t) = C_{CO_2,0}e^{-\frac{\dot{m}}{V}t} + \left(C_{ext} + \frac{r \cdot n}{\dot{m}}\right) \left(1 - e^{-\frac{\dot{m}}{V}t}\right) \quad (4)$$

Here, $C_{CO_2}(t)$ denotes indoor CO₂ concentration at time t , C_{ext} is outdoor CO₂, n is the number of occupants, r is the CO₂ generation rate per person, $\dot{m}$ is airflow rate, and V is room volume. This equation was used to simulate CO₂ trends and compare them with empirical observations, especially during periods of known occupancy. Consistency between predicted and measured curves validated the suitability of CO₂ for occupancy inference.

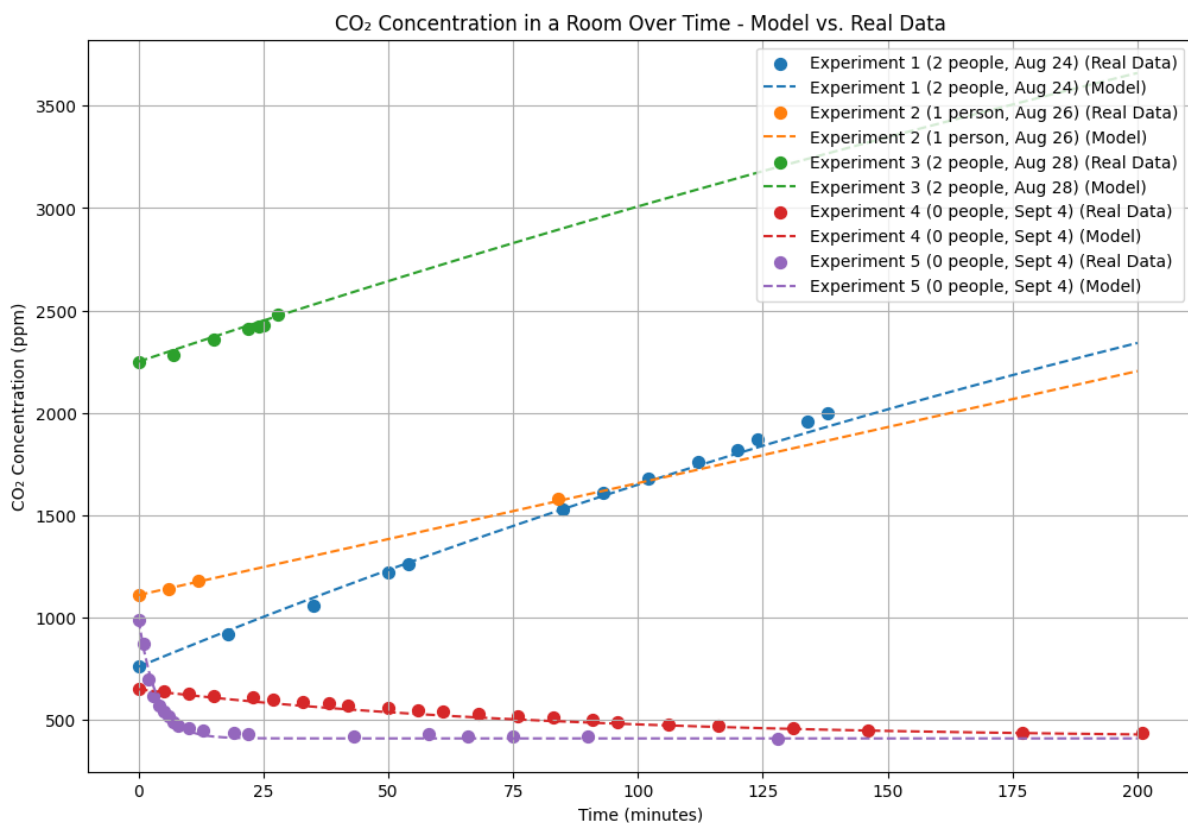


Рис. 11: Comparison of measured CO₂ concentration with theoretical predictions for varying occupancy levels.

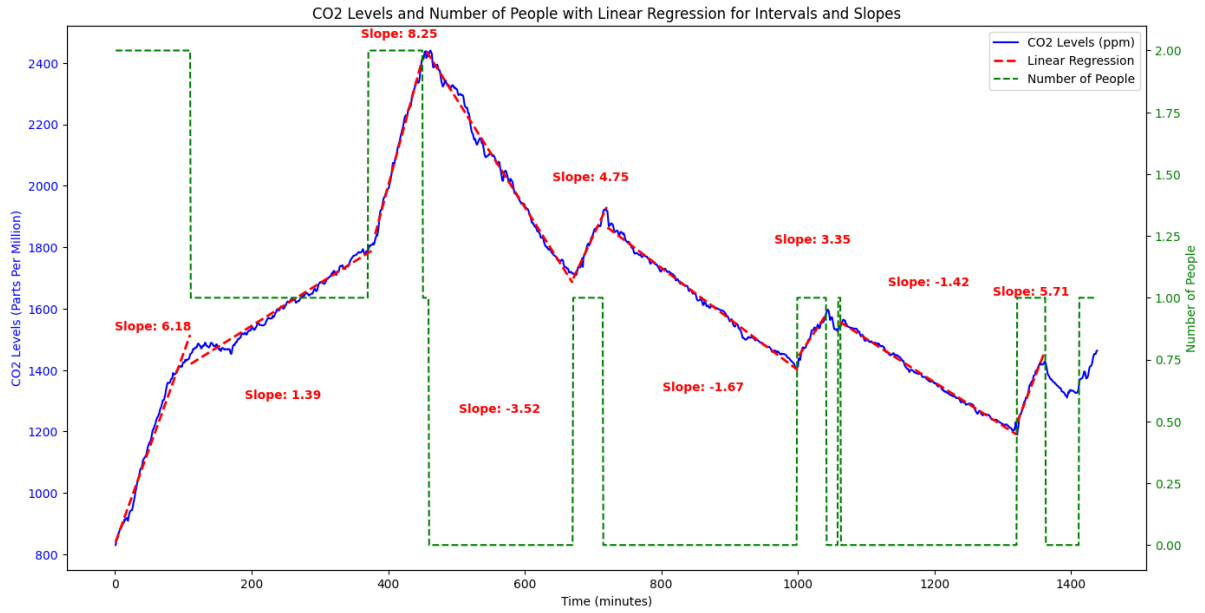


Рис. 12: CO₂ levels and number of people with linear regression for intervals and slopes.

5.4. Model Algorithms

5.4.1. k-Nearest Neighbors (k-NN)

The k-Nearest Neighbors (k-NN) algorithm is a classical non-parametric method used for classification and regression. It operates on a simple assumption: similar data points exist in close proximity in the feature space. For a new input sample, the algorithm identifies the k most similar samples in the training set and assigns the label based on a majority vote among them.

In this study, k-NN was employed to classify occupancy levels based on features derived from environmental sensor data. Each instance represented a 10-minute aggregated window containing statistical features such as CO₂ mean and slope, average temperature and humidity, and door state activity. Prior to classification, all features were min-max normalized to ensure uniform influence on the Euclidean distance calculation.

The optimal value of k was determined through grid search using 5-fold cross-validation, with $k = 2$ yielding the best performance in terms of F1-score across all classes.

Figure 13 illustrates the decision-making process of k-NN in a three-dimensional feature space. Test points are marked using colored crosses, while their two nearest neighbors are highlighted using yellow markers. The classification outcome depends on the majority label among these neighbors. For example, a test point surrounded by two class-1 samples is assigned to class 1, aligning with intuitive spatial reasoning.

While k-NN is conceptually simple and requires no training phase, it is computationally intensive during inference since all distances must be calculated at runtime. Moreover, its performance is sensitive to the scale of input features and to the density and distribution of training data in the feature space.

This makes proper preprocessing and feature engineering essential for applying k-NN effectively in occupancy detection tasks. Despite its simplicity, k-NN serves as a valuable baseline, offering interpretability and fast deployment for low-resource environments.

Summary of characteristics:

- *Advantages:* Highly interpretable, no training time, naturally supports multi-class output.
- *Limitations:* Sensitive to data scaling and sparsity; high inference cost for large datasets.

KNN 3D Visualization (K=2) - Binary Test Points

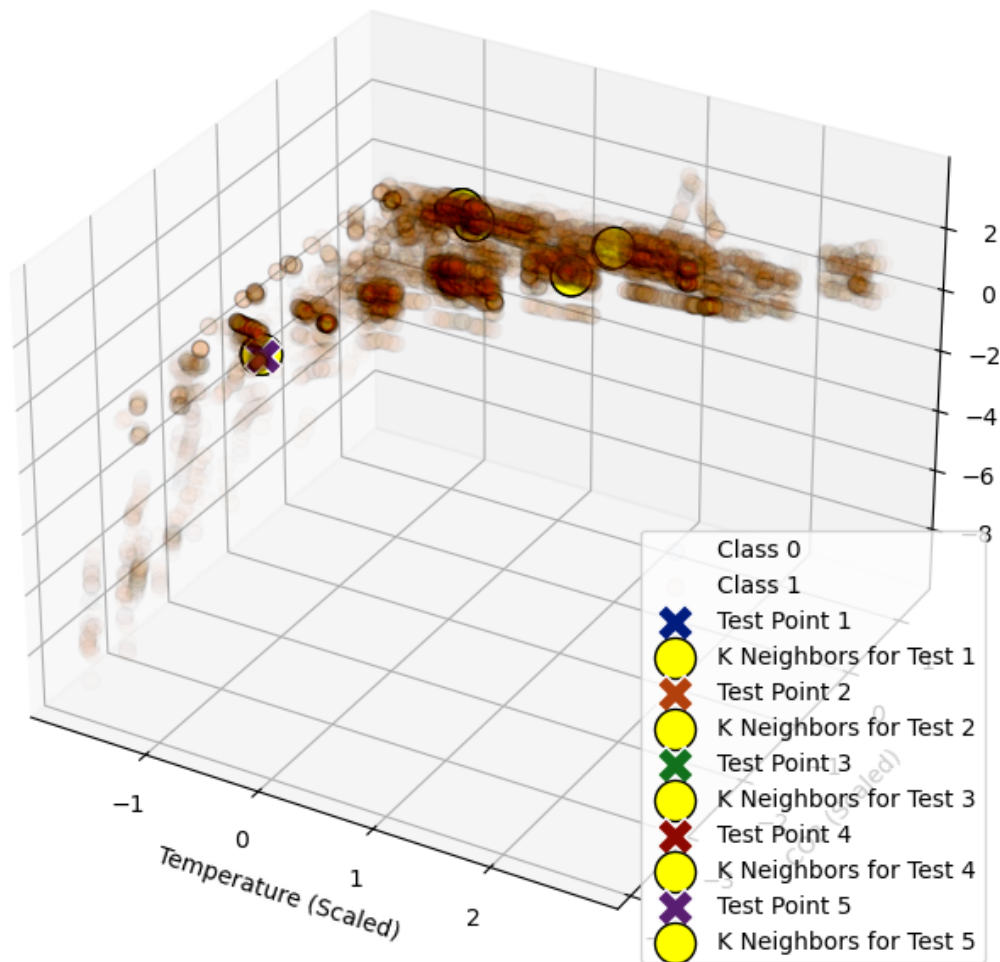


Рис. 13: 3D visualization of k-NN classification ($k = 2$) using test points and their nearest neighbors. Feature axes include scaled temperature, CO_2 , and its rate of change.

5.4.2. Random Forest (RF)

Random Forest is an ensemble learning algorithm based on decision trees. It constructs multiple trees during training and aggregates their outputs through majority voting, significantly reducing overfitting compared to individual trees. Each tree is built from a random bootstrap sample of the data, and at each node split, only a random subset of features is considered,

promoting model diversity.

In this study, Random Forest was used to classify occupancy levels based on derived features from CO₂, temperature, humidity, and door state signals. Each data point represents a temporal snapshot of indoor environmental conditions, encoded with rolling statistics and rate-of-change features that help reveal occupancy dynamics.

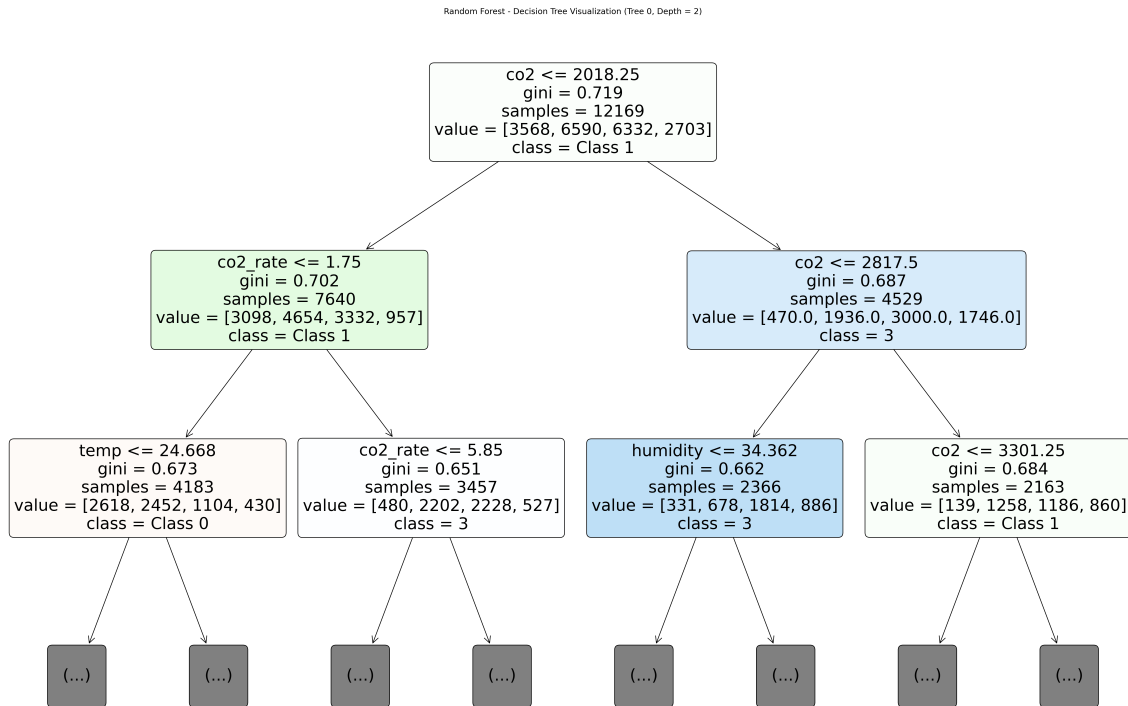


Рис. 14: Example of a decision tree from the trained Random Forest. Internal nodes contain splitting conditions based on environmental parameters (e.g., CO₂ thresholds). Leaf nodes represent occupancy class predictions.

Model configuration:

- Number of estimators (trees): 100
- Maximum depth: 16
- Random state: 42 (to ensure reproducibility)

These hyperparameters were selected based on empirical tuning to balance accuracy and overfitting. A depth of 16 allowed each tree to capture sufficient interaction between features while still generalizing across different occupancy profiles. The ensemble size of 400 trees ensured model stability without incurring excessive computational cost.

Advantages in this context:

- Handles non-linear relationships and feature interactions naturally
- Robust to outliers and noise in sensor data

- Provides internal metrics like feature importance, which aid in model interpretability

Despite its ensemble nature, Random Forests remain interpretable through techniques like individual tree visualization and feature importance ranking. This makes them suitable not only for performance but also for transparency in applications like building management systems, where explainability of predictions can be valuable.

Overall, Random Forest served as a strong classical benchmark in this study, offering reliable performance without extensive parameter tuning or sequence modeling.

5.4.3. XGBoost and LightGBM

While Random Forests rely on bagging—training multiple trees in parallel on different data subsets—XGBoost and LightGBM adopt a fundamentally different strategy known as boosting. In boosting, trees are built sequentially, with each new tree trained to reduce the residual errors of the ensemble built so far. This makes these models more expressive and generally more accurate for structured, tabular data.

XGBoost, short for Extreme Gradient Boosting, builds trees in a level-wise manner. It incorporates both L1 (Lasso) and L2 (Ridge) regularization terms into its objective function to help prevent overfitting. Additionally, it supports parallelization during tree construction, handles missing values natively, and implements second-order optimization, making it both flexible and efficient.

LightGBM, developed by Microsoft, improves on XGBoost in terms of training speed and memory usage. Rather than growing trees level-by-level, it uses a leaf-wise approach: it identifies the leaf with the maximum potential to reduce loss and splits it first. This often leads to deeper, more specialized trees. LightGBM also implements histogram-based decision rules, which bin continuous feature values and further reduce computational cost, especially for large datasets.

In this study, both XGBoost and LightGBM were trained on the same engineered features used for Random Forest. These features included statistical summaries, temporal derivatives, and contextual indicators extracted from CO₂, temperature, humidity, and door state measurements. Model training and evaluation followed the same preprocessing and split procedures used for other classical models.

The key hyperparameters for both models were selected through grid search. Each model used 400 estimators and a maximum depth of 20. The learning rate was set to 0.1, and a subsample ratio of 0.8 was used to promote generalization. For XGBoost specifically, a column subsample ratio of 0.9 was also applied to introduce additional randomness during feature selection.

Unlike Random Forests, which reduce variance by averaging multiple trees, XGBoost and LightGBM reduce both bias and variance by focusing learning on the hardest-to-predict samples. Although boosting models are typically more sensitive to hyperparameters and prone to overfitting if not tuned carefully, they tend to outperform bagging models in terms of raw accuracy. While Random Forests offer more straightforward interpretability through decision path tracing and feature importance, boosting models trade some of that interpretability for improved performance

and finer modeling of complex feature interactions.

In the context of this study, both boosting algorithms demonstrated high sensitivity to subtle shifts in CO₂ slopes and rolling humidity trends. Their ability to model fine-grained temporal and non-linear feature interactions made them particularly effective for intermediate occupancy levels, where signal patterns are more ambiguous. Moreover, both models handled missing values robustly, which is crucial in real-world deployments where sensor data may experience occasional dropout.

Although XGBoost and LightGBM are inherently less transparent than Random Forests, they significantly enhanced predictive performance in the multiclass classification task. Their contributions were especially notable in boundary cases where classical models struggled to differentiate between low and moderate occupancy. Overall, the use of boosting models added modeling depth to this study and served as a bridge to more sophisticated time-aware architectures such as LSTM.

5.4.4. Long Short-Term Memory (LSTM)

Traditional machine learning models treat each input instance as a static, independent sample. While this approach can yield acceptable results when features are carefully engineered, it fails to capture the underlying temporal dynamics that characterize real-world sensor data. In indoor environments, parameters like CO₂ concentration, temperature, and humidity evolve over time due to occupant behavior and ventilation effects. To learn from these patterns, a sequence-aware deep learning model is required. In this study, a Long Short-Term Memory (LSTM) network was employed to model these temporal dependencies.

LSTM is a type of recurrent neural network (RNN) specifically designed to learn from ordered sequences. Its architecture includes memory cells that are regulated by three types of gates: the input gate, forget gate, and output gate. These gates control how much of the previous memory is retained, how much new input is incorporated, and what information is output at each time step. This structure allows the LSTM to learn long-term dependencies in time-series data, unlike vanilla RNNs which suffer from vanishing gradients. The ability of LSTMs to “remember” relevant information across time steps makes them ideal for detecting occupancy patterns where environmental response is often delayed or prolonged.

Input Representation and Network Architecture The input to the LSTM model was formatted as a rolling window of 60 time steps, corresponding to 60 minutes of past sensor readings. Each time step contained three environmental features: CO₂ concentration, temperature, and relative humidity. This yielded a three-dimensional input tensor of shape (batch_size, 60, 3) for each training pass. The architecture consisted of two stacked LSTM layers, each with 64 hidden units. After the first LSTM layer, a dropout regularization layer with a dropout rate of 0.5 was applied to reduce overfitting. The final hidden state was fed into a dense output layer with a softmax activation function to classify the input sequence into one of four occupancy levels: 0, 1, 2, or 3+ people.

Batch Size The model was trained using a mini-batch size of 32. A batch is a subset of the training data passed through the model before updating the weights. Using smaller batches (e.g., 32) allows for more frequent updates and helps the model converge faster, especially when

memory constraints are not an issue. It also introduces a level of stochasticity which can help escape local minima. In this study, a batch size of 32 was empirically selected to balance training speed with convergence stability.

Optimizer and Learning Rate Model optimization was carried out using the Adam optimizer, which combines the benefits of two other extensions of stochastic gradient descent: AdaGrad and RMSProp. Adam is particularly suitable for problems with sparse gradients and noisy data, characteristics common in sensor-based occupancy estimation. The learning rate was set to 0.002. This hyperparameter controls the step size in parameter space during gradient descent. A smaller learning rate ensures more stable convergence and reduces the risk of overshooting minima in the loss landscape. The combination of Adam with a moderate learning rate provided both stability and efficiency.

Loss Function and Class Imbalance Handling To accommodate the multi-class nature of the problem, categorical cross-entropy was used as the loss function. This loss measures the divergence between the predicted probability distribution and the actual class labels. However, since the dataset exhibited a degree of class imbalance — with unoccupied and single-occupancy states appearing more frequently — a weighted version of cross-entropy was implemented. This adjustment penalized misclassifications of underrepresented classes more heavily, thereby encouraging the model to improve its sensitivity to rare classes such as ‘3+ occupants’.

Early Stopping and Epochs The model was trained for a maximum of 300 epochs, where each epoch represents one complete pass through the training dataset. To avoid overfitting, early stopping was employed with a patience value of 10. This means that if the model's validation performance did not improve for 10 consecutive epochs, training was terminated. Early stopping not only conserves computational resources but also ensures that the model generalizes well to unseen data.

Train-Test Split and Cross-Validation The dataset was initially divided into training and testing subsets using an 80/20 stratified split, ensuring that the occupancy classes were proportionally represented in both partitions. To further validate the model's robustness, 10-fold cross-validation (CV) was employed. In k -fold CV, the data is partitioned into k equal parts. The model is trained k times, each time using a different fold as the validation set and the remaining $k - 1$ folds for training. This technique reduces the likelihood that performance metrics are overly optimistic due to favorable data splits. In this study, 10-fold CV helped assess the model's generalization capability and ensured that the reported accuracy and F1-scores were not biased by a single partitioning.

Summary of Configuration Overall, the LSTM model was configured to learn directly from raw sequential data with minimal feature engineering. The combination of sequence modeling, regularization, stratified data splitting, and cross-validated performance evaluation resulted in a model that not only achieved high classification accuracy but also maintained robustness across folds. Its ability to track the dynamic response of indoor environmental variables made it especially effective in occupancy detection based on CO₂ and other ambient factors.

6. Results

This section presents the performance evaluation of the proposed CO₂-based occupancy detection framework. The experiments are divided into binary and multi-class classification tasks, covering both classical machine learning models and the LSTM-based deep learning model.

6.1. Experiment 1: Binary Occupancy Classification

The first experiment focused on binary classification to determine whether a room was occupied (1) or vacant (0). This task is directly relevant to HVAC optimization scenarios, where systems can switch off ventilation or heating/cooling when the room is empty to conserve energy.

Four classical machine learning models were evaluated: XGBoost, LightGBM, Random Forest, and k-Nearest Neighbors (KNN). The models were trained on preprocessed sensor features including CO₂ concentration, temperature, humidity, and ventilation indicators.

Таблица 2: Binary classification performance of classical machine learning models

Model	Accuracy	F1 Score	Precision	Recall
XGBoost	95.99%	97.68%	96.87%	98.49%
LightGBM	95.24%	97.26%	95.83%	98.74%
Random Forest	96.11%	97.75%	96.81%	98.72%
KNN (k=2)	88.05%	93.17%	91.31%	95.10%

These results indicate that CO₂ and environmental signals provide a strong basis for distinguishing occupied from unoccupied states. While tree-based models (XGBoost, LightGBM, RF) yielded nearly identical high performance, KNN lagged slightly due to its local decision boundaries.

Confusion Matrices:

The confusion matrices for each model (Figures 15 to 18) offer deeper insight into classification behavior.

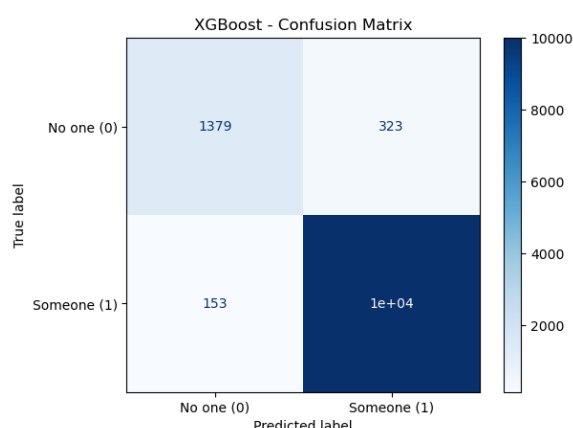


Рис. 15: Binary classification confusion matrix for XGBoost.

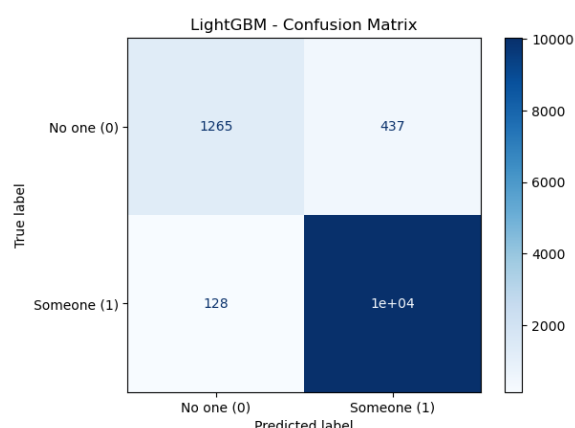


Рис. 16: Binary classification confusion matrix for LightGBM.

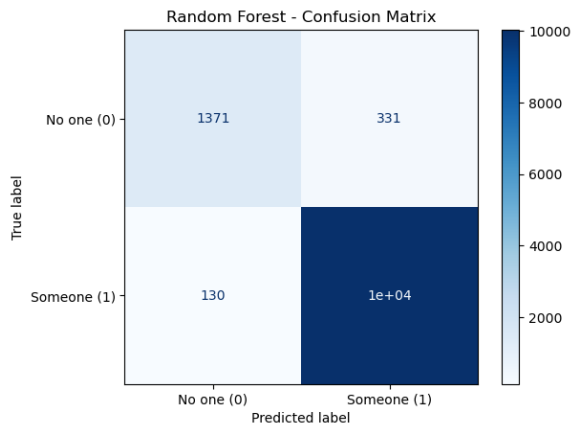


Рис. 17: Binary classification confusion matrix for Random Forest.

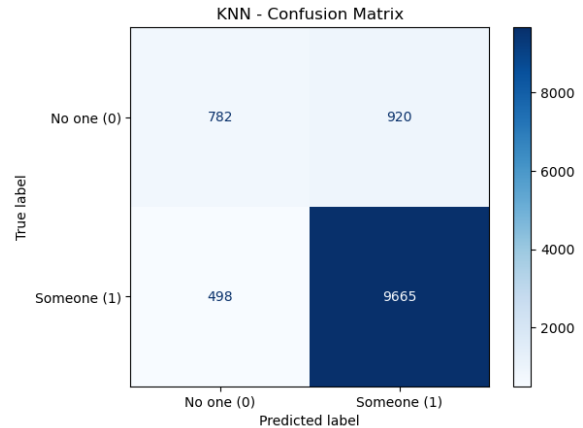


Рис. 18: Binary classification confusion matrix for k-Nearest Neighbors (k=4).

These matrices confirm that the majority of classification errors involved occasional false positives—predicting presence during vacancy—likely due to transient environmental noise. Nonetheless, high precision and recall values for all models demonstrate that short-term CO₂ and environmental trends are highly informative for real-time binary occupancy inference.

6.2. Experiment 2: Multi-Class Occupancy Classification

In the second experiment, we extended the classification task to a multi-class problem, aiming to predict the exact number of occupants in the room (ranging from 0 to 3). This task is especially valuable for fine-tuning HVAC control systems by allowing proportional adjustments to ventilation or conditioning based on precise occupancy levels, thereby maximizing energy efficiency and comfort.

We evaluated the same set of classical machine learning models: XGBoost, LightGBM, Random Forest, and K-Nearest Neighbors (KNN). Each model was trained on the same feature set, including CO₂ concentration and its rate of change, temperature, humidity, and door/window status indicators.

Performance Summary:

[HTML]F3F315 Model	Accuracy	F1 Score (Weighted)
XGBoost	0.87	0.87
LightGBM	0.87	0.87
KNN	0.86	0.86
Random Forest	0.87	0.87

Таблица 3: Performance of classical models on multi-class occupancy classification.

Despite having a slight class imbalance (with more frequent occurrences of 1 and 2 occupants), all models demonstrated strong generalization. XGBoost and LightGBM slightly outperformed Random Forest and KNN in distinguishing among the four occupancy states, particularly during transitions and high occupancy counts.

Confusion Matrices:

To further illustrate model behavior, we present the confusion matrices for Random Forest and XGBoost (Figures 19 and 20).

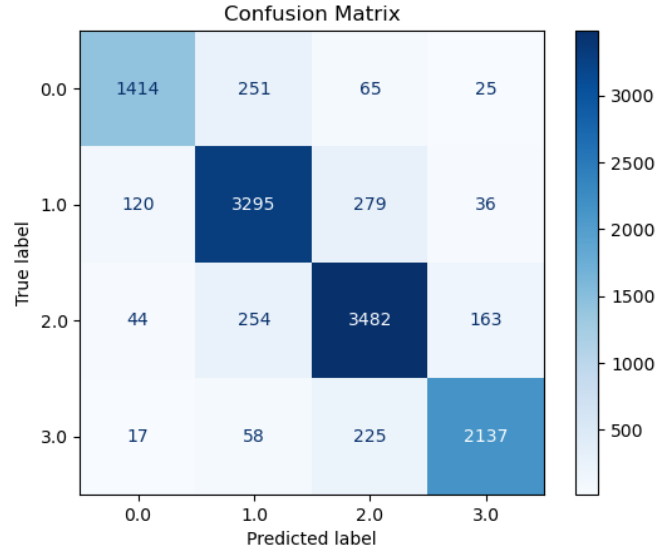


Рис. 19: Multi-class confusion matrix for Random Forest.

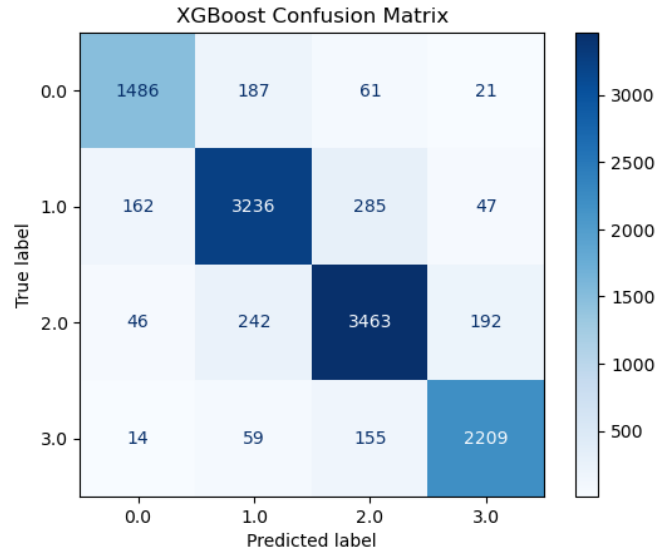


Рис. 20: Multi-class confusion matrix for XGBoost.

Both models show high accuracy for classes 1 and 2, which form the majority of the dataset. The slight misclassifications between adjacent classes (e.g., class 2 predicted as class 1 or 3) are expected due to the overlapping sensor patterns during transitional periods. Nonetheless, the consistent diagonal dominance in these confusion matrices confirms reliable performance across all occupancy states.

6.3. Experiment 3: Deep Learning with LSTM

To capture the temporal dependencies in sensor signals, we deployed a two-layer Long Short-Term Memory (LSTM) neural network for multi-class occupancy classification (0 to 3 occupants). Unlike classical models that rely on independently sampled feature vectors, the LSTM leverages sequential input windows (60-minute time steps) of CO₂, temperature, and humidity, enabling it to learn from patterns of change over time.

The LSTM model achieved a test accuracy of **97%** and a weighted F1-score of **97%**, outperforming all classical machine learning models in the same task. Notably, these results were obtained without applying any resampling techniques such as SMOTE or data balancing. This indicates that the model was able to generalize well despite potential class imbalance, likely due to the strong temporal patterns embedded in the environmental signals.

These metrics confirm that incorporating temporal structure into the learning process significantly enhances the ability to infer occupancy. The LSTM's ability to contextualize current sensor values based on historical behavior makes it especially robust to transient disturbances and short-lived fluctuations in CO₂ or temperature that may mislead static classifiers.

6.3.1. Confusion Matrix Analysis

The confusion matrix for the LSTM model (Figure 21) provides a detailed breakdown of classification performance across all four occupancy levels. The diagonal dominance indicates strong prediction accuracy for each class, with minimal misclassification between adjacent occupancy levels.

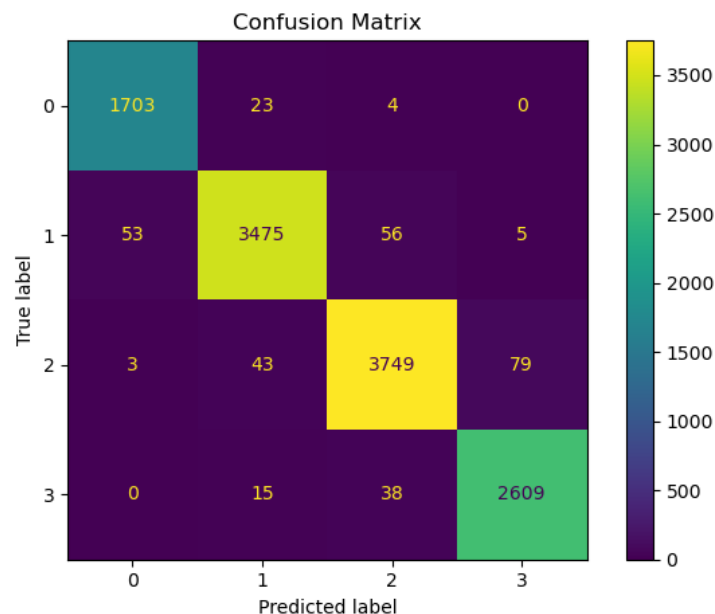


Рис. 21: Confusion matrix for LSTM multi-class occupancy prediction (0–3 occupants).

Most of the classification errors are limited to neighboring classes—e.g., occasional confusion between 1 and 2 occupants or between 2 and 3 occupants. This is expected in real-world scenarios where transient sensor behaviors can blur the boundary between closely related occupancy states.

For instance, periods when occupants briefly enter or leave the room may result in ambiguous environmental signals, making it challenging for even a sequence-based model to detect abrupt transitions instantly. Nonetheless, the LSTM model maintains high precision and recall across all classes, suggesting reliable multi-class discrimination.

The strong performance on classes 2 and 3, which are typically more challenging due to overlapping CO₂ ranges, demonstrates the LSTM's effectiveness in modeling cumulative environmental effects rather than relying on static thresholds.

6.3.2. Training and Validation Loss Curve

Figure 22 illustrates the training and validation loss curves of the LSTM model over 300 epochs. The plot shows a smooth and consistent decrease in both training and validation loss, reflecting stable convergence and effective generalization.

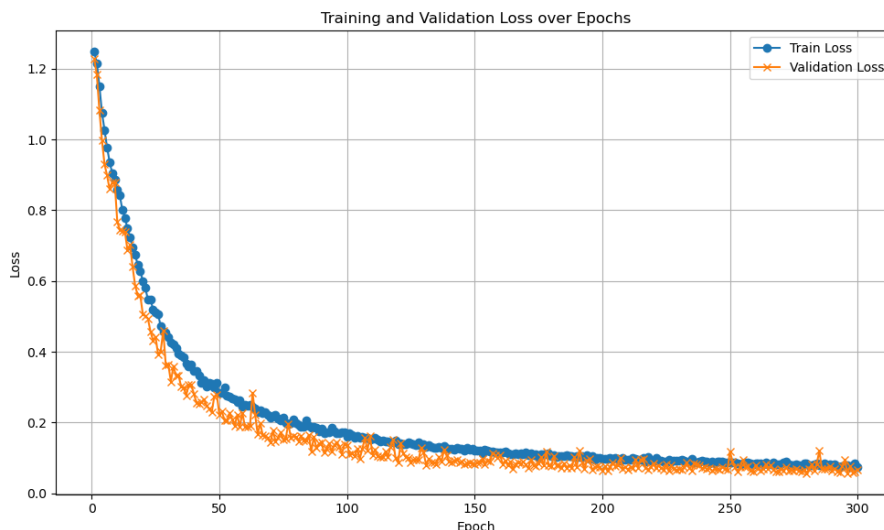


Рис. 22: Training and validation loss curves for the LSTM model over 300 epochs.

The validation loss closely follows the training loss, with no signs of overfitting throughout the training process. Early fluctuations in the first 50 epochs gradually stabilize, indicating the model's increasing ability to minimize sequence-level prediction errors.

By the 100th epoch, the model reaches a plateau in both curves, which is typically a sign that the optimizer has found a well-generalizing minimum. The learning behavior supports the effectiveness of the selected architecture, training configuration (e.g., Adam optimizer, learning rate of 0.002), and regularization strategy (dropout rate of 0.5).

This convergence behavior confirms the robustness of the LSTM architecture in capturing temporal dependencies in environmental signals such as CO₂, temperature, and humidity for occupancy classification.

6.3.3. Temporal Prediction vs Ground Truth

To qualitatively evaluate the LSTM model's behavior in real time, Figure 23 presents a time-aligned comparison of predicted occupancy (dashed red) versus the ground truth (black), plotted alongside CO₂ concentration (blue) across a 16-hour segment. This visualization offers insight into how the model responds to transitions in human presence, as well as its robustness against short-term noise.

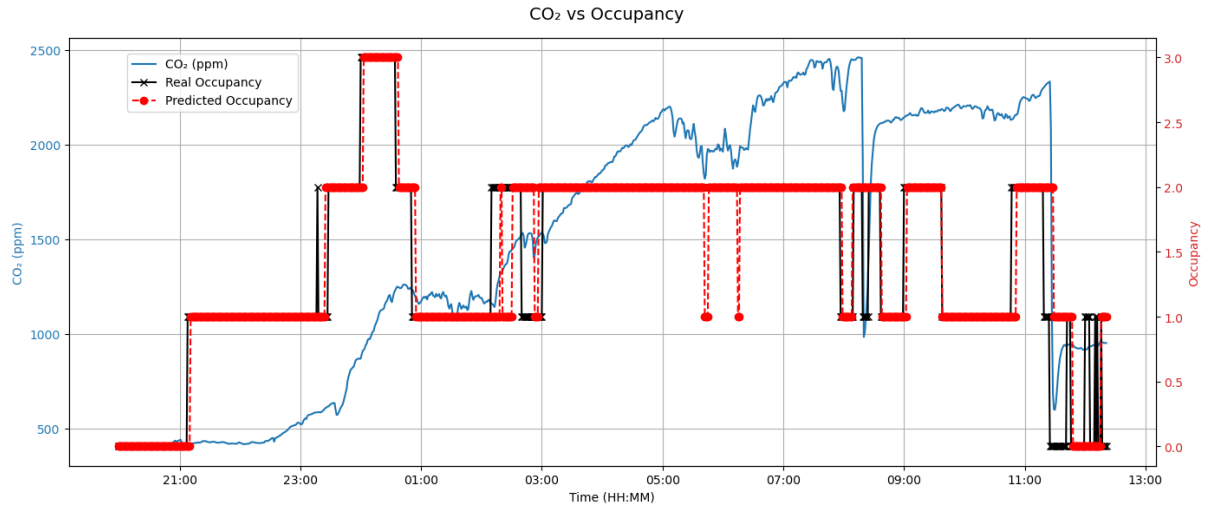


Рис. 23: Comparison of CO₂ levels (blue), ground truth occupancy (black), and LSTM-predicted occupancy (dashed red) over a 16-hour window.

The figure demonstrates that the LSTM model effectively tracks occupancy changes with high temporal resolution. Most occupancy transitions—such as arrivals, departures, and occupant count increases—are accurately identified, with predicted values closely following the true labels throughout the day. The predicted curves exhibit minimal jitter, suggesting that the model has learned to generalize over short-term fluctuations in environmental signals.

However, slight temporal lags are visible between the ground truth and predictions, especially during sharp transitions. These short delays are physically expected and are not necessarily modeling failures. They arise because CO₂ concentration is an indirect indicator of presence that requires time to change in response to human activity. When a person enters the room, the CO₂ they exhale does not instantaneously raise the ambient level; rather, it disperses through the air volume, mixes with existing concentration, and reaches the sensor location after a brief delay. Similarly, when occupants exit, it takes time for CO₂ to decay or ventilate depending on airflow dynamics. These delays typically span a few minutes, depending on room size, sensor placement, and ventilation rate.

The LSTM model appears to have learned these temporal response patterns. For example, in multiple segments around 23:00–01:00 and again between 09:00–11:00, the model prediction trails the ground truth by a short margin, corresponding exactly to expected CO₂ diffusion delays. This behavior highlights the benefit of using a sequence model: rather than making frame-by-frame predictions, the LSTM relies on trends and recent context, smoothing over brief ambiguities.

Moreover, the model avoids erratic class switching and shows resilience during flat CO₂ periods, maintaining stable occupancy estimates even when changes are subtle. This consistency is crucial for deployment in HVAC control systems, where unnecessary toggling can cause inefficiencies.

In summary, the LSTM not only achieves numerical accuracy but also captures the underlying temporal structure of occupancy transitions. Its predictions exhibit smoothness, responsiveness, and physical plausibility, making it well-suited for applications where real-time and delay-tolerant inference are both necessary.

7. Discussion

7.1. Overview of Findings

The dataset collected for this study captured a realistic distribution of occupancy levels within a residential environment. As shown in Figure 24, the majority of recorded time windows involved one or two occupants (31.4% and 33.2%, respectively), while fully unoccupied periods accounted for 14.3% and fully occupied instances (three people) made up 21.1%. This natural variability in occupancy states provided a meaningful and balanced foundation for training and evaluating both classical and deep learning models.

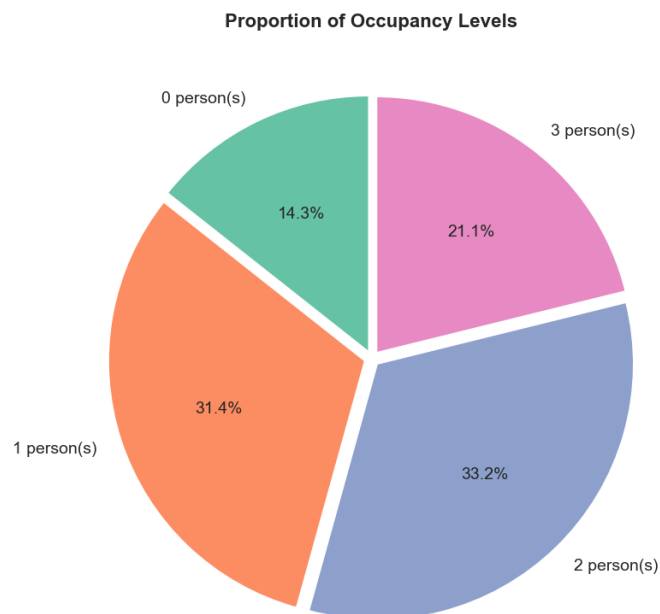


Рис. 24: Proportion of occupancy levels across the dataset. The most common states were one-person and two-person occupancy, representing nearly two-thirds of the total data.

This study aimed to evaluate the efficacy of CO₂-based occupancy detection models in residential environments using both classical machine learning algorithms and deep learning methods, particularly Long Short-Term Memory (LSTM) networks. Across multiple experiments, the results consistently demonstrated that incorporating temporal dependencies in sensor signals substantially improves occupancy detection performance. Notably, the LSTM model achieved an accuracy and F1-score of 97% in the multi-class classification task (0–3 occupants), significantly outperforming traditional models like Random Forest, XGBoost, and k-Nearest Neighbors, even without data balancing techniques.

7.2. Interpretation of Model Performance

The classical machine learning models benefited from strong preprocessed feature sets, particularly those derived from CO₂, temperature, and humidity gradients. However, their performance plateaued in capturing the sequential dependencies inherent in occupancy behavior. The LSTM, in contrast, effectively learned these temporal dynamics by processing one-hour sequential windows, allowing it to model delayed CO₂ accumulation and dissipation patterns associated with occupant presence.

Interestingly, the LSTM performed robustly even without synthetic data augmentation (e.g., SMOTE), suggesting that the temporal structure alone enabled it to generalize well, even under class imbalance. This underscores a key insight: time-series patterns in environmental data inherently encode occupancy states in a way that static models cannot fully exploit.

7.3. Comparison with Existing Literature

Most prior studies in CO₂-based occupancy detection have focused on controlled environments such as offices, classrooms, or conference halls, where CO₂ generation is more predictable. In contrast, this study contributes novel insights by applying the methodology to a residential setting—a more dynamic and complex environment. Compared to related works, such as those using Markov Switching Models or linear regression over CO₂ levels, the LSTM-based approach demonstrated superior granularity and adaptability.

Moreover, unlike simulation-based studies which often rely on idealized assumptions about ventilation rates and occupant behavior, this work leverages real sensor data collected over several months. This grounds the results in practical conditions, increasing their reliability and applicability in real-world residential contexts.

7.4. Implications for Smart HVAC Control

The high accuracy of occupancy estimation using LSTM models has significant implications for smart HVAC systems. Traditional HVAC systems often operate on predefined schedules or simple motion sensor triggers. By integrating CO₂-based occupancy inference, HVAC systems can dynamically adapt their operation based on real-time occupant presence, leading to substantial energy savings without compromising thermal comfort.

In addition to energy optimization, this study also highlights the importance of maintaining indoor air quality. As shown in Figure 25, more than half of the measured CO₂ readings (54.1%) exceeded 1400 ppm—classified as poor air quality. Only 9.9% of the samples fell within the “Excellent” range (≤ 600 ppm). This skewed distribution emphasizes the inadequacy of conventional ventilation systems in meeting comfort thresholds under real occupancy conditions.

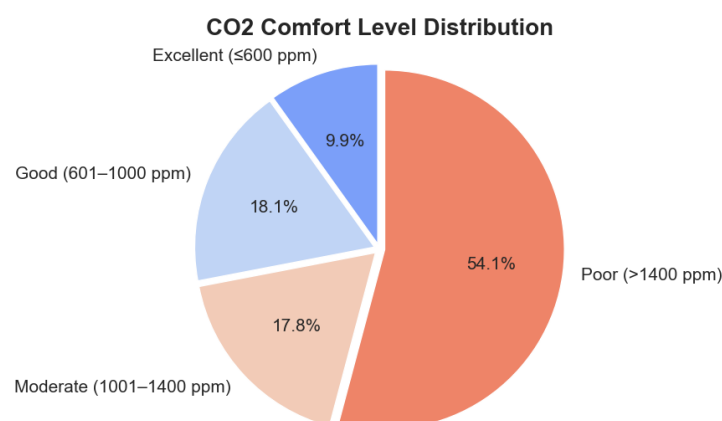


Рис. 25: CO₂ comfort level distribution based on indoor sensor data. The majority of values exceeded recommended thresholds, indicating frequent periods of poor indoor air quality.

These results reinforce the need for intelligent, demand-responsive ventilation strategies that are aware of both occupant presence and indoor pollutant accumulation. Occupancy-driven

control frameworks—especially those informed by CO₂ dynamics—can therefore play a dual role: improving energy efficiency and ensuring healthier indoor environments.

7.5. Limitations

Despite the promising findings, several limitations must be acknowledged. The dataset used for model development was collected exclusively from a residential building environment. While it spans several months and incorporates a variety of environmental sensors, its applicability to other building types—such as commercial, institutional, or industrial settings—remains to be evaluated. Each of these contexts may exhibit distinct occupancy patterns, ventilation systems, and space utilization behaviors that could influence model performance.

Although LSTM models demonstrated strong predictive capabilities, they are computationally demanding and require careful tuning of architecture and hyperparameters. This can pose challenges for deployment in resource-constrained environments, such as embedded HVAC controllers or edge devices.

Additionally, certain internal activities—such as cooking—can affect CO₂ concentrations independently of occupancy, potentially introducing noise into the signal. These behavioral variables were not explicitly labeled or modeled in this study. Furthermore, modeling occupancy in multi-zone residential layouts introduces added complexity, as CO₂ dynamics vary significantly between interconnected rooms due to airflow and diffusion delays. These challenges highlight the need for more granular sensor deployment and spatial modeling in future research.

7.6. Future Directions

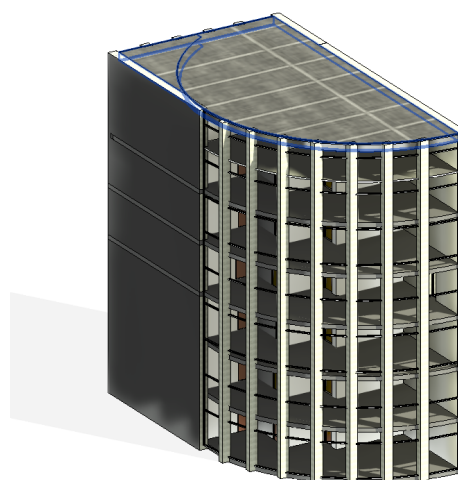
To further enhance CO₂-based occupancy detection and integrate it into intelligent HVAC control systems, several promising directions are proposed. The increasing demand for energy-efficient residential buildings, combined with the availability of low-cost environmental sensors and simulation tools, makes this line of research both timely and impactful.

A foundational step in this progression involves the development of a high-fidelity digital twin capable of simulating occupancy behavior, airflow patterns, and HVAC dynamics in detail. Digital twins bridge the gap between physical and virtual systems, enabling real-time monitoring, performance evaluation, and control strategy optimization. In this study, preliminary efforts toward such a twin were made through the detailed 3D modeling of a six-story residential building—the Skoltech dormitory.

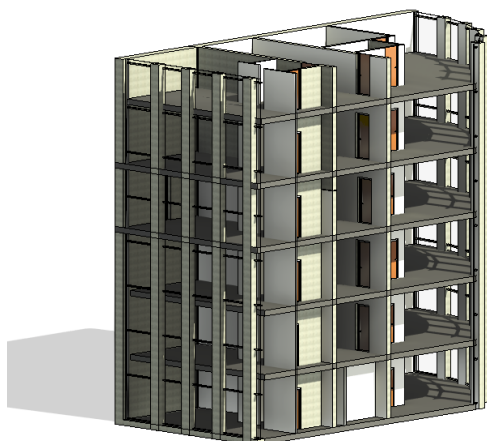
Using Autodesk Revit, a comprehensive Building Information Model (BIM) was constructed, capturing architectural and spatial zoning details. This digital environment provides a scalable framework for mapping thermal zones, CO₂ sensors, and HVAC elements, forming the basis for simulation in tools like EnergyPlus or Modelica. As shown in Figure 26, various views of the building illustrate the level of detail incorporated, including cross-sections, exterior perspectives, and internal partitioning.



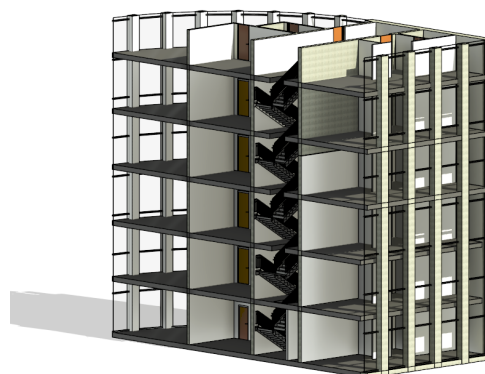
Top-left: Vertical cross-section



Top-right: Full exterior perspective



Bottom-left: Internal partitions

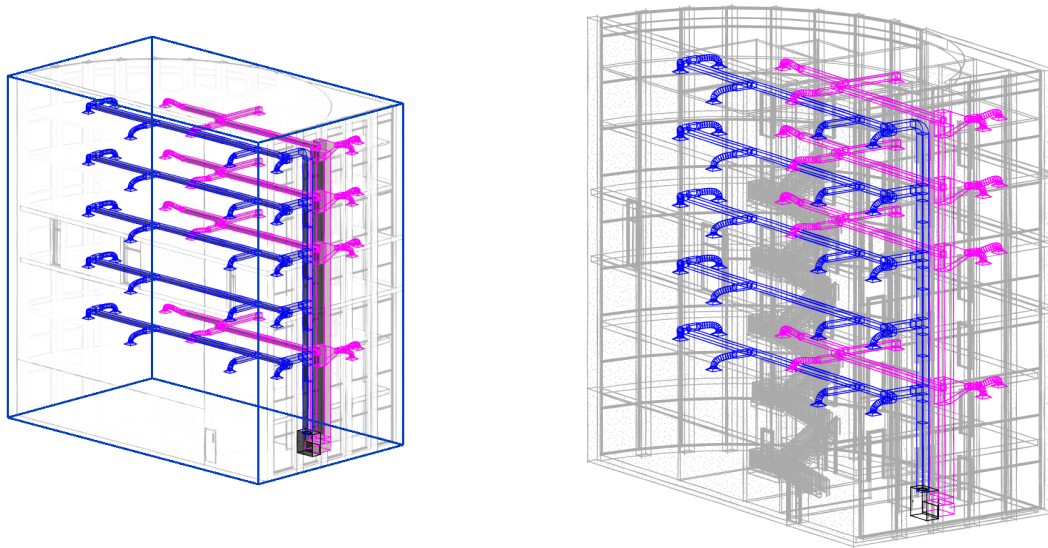


Bottom-right: Stairwell shaft

Рис. 26: Revit-based 3D modeling of the six-story Skoltech dormitory building. The views illustrate (top-left) a vertical cross-section showing room layouts and corridors, (top-right) a full exterior perspective with curved façade, (bottom-left) a semi-exposed view of internal partitions, and (bottom-right) the stairwell shaft and structural segmentation. These visualizations support future digital twin development and HVAC system simulation.

Building on this model, a complete HVAC system was virtually installed to demonstrate zone-level ductwork and ventilation routing. Figure 27 presents the mechanical system layout, with color-coded supply (blue) and return (magenta) ducts spanning all floors. These visualizations

form the basis for implementing and testing occupancy-driven control logic in a simulated environment.



Left: Simplified view with transparent shell and HVAC ducts highlighted.

Right: Full structural overlay showing duct integration within the architectural frame.

Рис. 27: 3D visualization of the HVAC system installation across the six-story Skoltech dormitory building. Supply ducts (blue) and return ducts (magenta) are routed through all floors, terminating at diffusers within each zone. These views demonstrate the mechanical layout designed in Revit, supporting future simulation and control development.

Recent literature underscores the growing importance of integrating occupancy prediction models into HVAC control frameworks. Wang et al. [63] review a wide range of smart HVAC strategies and highlight that accurate occupancy forecasting—particularly when powered by machine learning—can significantly enhance energy efficiency and indoor comfort. However, they also note key barriers such as occupant unpredictability and data scarcity in residential settings. The LSTM-based occupancy model developed in this study addresses these challenges by learning from real-world sensor data, providing a practical foundation for intelligent control schemes.

With this infrastructure in place, the next phase will involve embedding the CO₂-based occupancy forecasting model into a real-time Model Predictive Control (MPC) system. Unlike traditional rule-based control, MPC enables anticipatory decision-making by leveraging occupancy forecasts to optimize ventilation, heating, and cooling schedules. This approach has the potential to significantly reduce energy consumption while maintaining comfort.

Future work will also explore hybrid modeling approaches that fuse physics-based knowledge (e.g., mass-balance CO₂ dynamics, switching Markov processes) with deep learning models such as LSTM. This fusion is expected to improve robustness in unpredictable ventilation scenarios while maintaining generalization to new environments.

Lastly, to ensure year-round reliability, extended data collection over multiple seasons will be conducted. Additionally, lightweight model variants will be developed for edge deployment on embedded controllers, enabling practical implementation in real residential settings.

Overall, these future directions aim to transform occupancy detection from a passive sensing task into a central component of adaptive, intelligent HVAC systems—bridging data-driven methods with control-oriented applications for sustainable residential building operation.

8. Conclusion

This thesis explored the development and evaluation of CO₂-based occupancy detection models for residential buildings, emphasizing their role in enabling intelligent and energy-efficient HVAC control. By comparing classical machine learning algorithms with temporal deep learning approaches, it was demonstrated that Long Short-Term Memory (LSTM) networks offer significantly higher accuracy and robustness in multi-class occupancy classification tasks, even without data augmentation.

The work contributes to the growing field of sensor-driven building intelligence by addressing the relatively underexplored domain of residential occupancy detection, where occupant behavior is less structured and more variable than in commercial settings. The integration of real-world, long-term environmental sensor data enhances the practical relevance and deployability of the proposed methods.

Beyond algorithmic development, this research established a digital foundation for future experimentation through the creation of a detailed BIM-based model of a six-story residential building and the installation of a virtual HVAC system. These assets support future efforts in digital twin development, simulation-based control testing, and integration into Model Predictive Control (MPC) frameworks.

In summary, this work lays the groundwork for transforming passive occupancy monitoring into an active control strategy, bridging data-driven inference with adaptive HVAC operation. The insights and infrastructure developed herein provide a strong platform for advancing sustainable, occupant-aware building technologies.

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